



Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM:
**PROJECTING LONG-TERM LUMINOUS,
PHOTON, AND RADIANT FLUX
MAINTENANCE OF LED LIGHT SOURCES**
AN AMERICAN NATIONAL STANDARD



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Publication of this Technical Memorandum
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
IES Testing Procedures Committee**



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An accompanying online calculator for ANSI/IES TM-21-21 is available: www.ies.org/tools

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1.0 Introduction and Scope

1.1 Introduction

One of the benefits that LED light sources can provide is very long usable life. Unlike other lighting technologies, LEDs typically do not fail catastrophically* during use. However, over time the light output will gradually depreciate. At some point in time the light emitted from an LED depreciates to a level where it is no longer considered adequate for a specific application. It is important in lighting design to understand when this “useful lifetime” of an LED source is reached.

ANSI/IES LM-80 defines the setup, conditions, and procedures for performing luminous flux maintenance testing of LED light sources and is the IES standard that is used widely to measure the lumen depreciation behavior of LEDs. LED device manufacturers routinely provide ANSI/IES LM-80 reports for their products, with data collected during testing for 6,000 hours or more. However, how the data collected from ANSI/IES LM-80 testing is used to best determine the useful lifetime of the tested product is not well defined.

The rated flux maintenance life of an LED is the elapsed operating time over which an LED light source maintains a given percentage of its initial light output. It is defined as τ_p , where τ is the lumen, photon or radiant flux, and p is the percentage value. For example, L_{70} is the time (in hours) when the luminous flux output from the LED has dropped to 70% of initial; Q_{90} is the time (in hours) when the photon flux output from the LED has dropped to 90% of initial; R_{80} is the time (in hours) when the radiant flux output from the LED has dropped to 80% of initial. The flux maintenance of an LED light source is dependent upon many variables, including the operating temperature, drive current, and technology and materials used to construct the product. As such, the flux maintenance of LEDs can vary not only from manufacturer to manufacturer, but also between different LED package types produced by a single manufacturer.

* Used in this sense, a “catastrophic failure” is a sudden failure that results in a complete inability to perform all required functions of an item. (Laplante, Philip A. Dictionary of Computer Science, Engineering and Technology. CRC Press, 2017)

This Technical Memorandum recommends a method of projecting the flux maintenance of LED light sources from the data obtained by ANSI/IES LM-80 testing.

This document was developed by a dedicated TM-21 Working Group of LED industry professionals. The analyses of the ANSI/IES LM-80-15 (the 2015 publication of ANSI/IES LM-80) test data provided by major LED manufacturers are used to rationalize and support this document. Much of the ANSI/IES LM-80-15 data came from testing that extended to 9,000 hours, and in some cases beyond.

Two additional notes are relevant here:

- In 2018, IES issued *PS-10-18, IES Position on LED Product Lifetime Prediction*.¹
- An accompanying online calculator for ANSI/IES TM-21-21 is available: www.ies.org/tools.

1.2 Scope

This document provides recommendations for projecting flux maintenance of LED light sources using data obtained when testing them per ANSI/IES LM-80.

This method shall not be used to project lumen, photon or radiant flux maintenance below 70%.

2.0 Normative Reference

Illuminating Engineering Society. *ANSI/IES LM-80-21, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules*. New York: IES; 2021.

3.0 Definitions

In addition to the terms defined in this section, illuminating engineering terms are defined in ANSI/IES LS-1-20.²

3.1 DUT

The *device under testing* is the LED light source as defined in **Section 3.2**.

3.2 LED light source

An LED package, array, or module that is operated via an auxiliary driver.*

3.3 rated flux maintenance life, τ_p

The elapsed operating time (hours) over which the LED light source will maintain the percentage, p , of its initial flux output*; e.g.:

- τ_{70} = time to 70% flux maintenance
- τ_{90} = time to 90% flux maintenance
- τ = L (luminous flux), Q (photon flux), or R (radiant flux)

4.0 Data and Sample Size

4.1 Data to Be Used

The data to be used in this projection method shall be collected according to the methods described in ANSI/IES LM-80, provided the measurement intervals comply with the uniformity requirements of **Section 4.3**.

4.2 Sample Size Recommendation

All data from the sample set at a given case temperature and drive current from the ANSI/IES LM-80 test report for a specific product model should be used for flux maintenance life projection. The recommended number for the sample set is a minimum of 20 units in order to be able to use a multiplication factor of 6 times the test duration, as specified in **Section 5.2.5**, for flux maintenance life projection.

A sample size of fewer than 20 units may be used when specified by the requestor of the application of this estimation method. When a sample size smaller than 20 units is specified and used, the sample size shall be noted in any reporting of the use of this method. Any change in sample size will lead to a change in uncertainty and the time interval for the projection of the luminous flux maintenance life. For a sample size of 10 units, a multiplication factor of 5.5 times the test duration, as specified in **Section 5.2.5**, shall be used for flux maintenance life projection.

4.3 Flux Data Collection

Additional measurements after the initial 1,000 hours at intervals less than 1,000 hours are encouraged. Additional measurements beyond 6,000 hours are encouraged and will provide the basis for more accurate flux maintenance predictions.

Data collected within 48 hours (a 96-hour window) of each “1,000 hour measurement point”—e.g., for a 1,000-hour interval, data collected from 952 hours to 1,048 hours, or from 1,952 hours to 2,048 hours—are acceptable for use in product flux maintenance projections. This 48-hour leeway is also applicable to intervals smaller than 1,000 hours. The exact time designation shall be used in the projection calculation.

The same time interval requirements shall apply to luminous flux maintenance, photon flux maintenance and radiant flux maintenance projections.

4.4 Sample Selection and Data Usage

The DUT samples tested per ANSI/IES LM-80 shall be selected to be sufficiently representative of the overall population of the product. All data collected for the DUT samples tested per ANSI/IES LM-80 shall be used for projection, with the exception of failed DUT samples as defined in ANSI/IES LM-80. Specially built samples to be used as DUTs shall not be allowed. Selectively excluding data based on performance shall not be allowed.

5.0 Flux Maintenance Life Projection

5.1 Method

The recommended method of flux maintenance projection is the application of a curve-fit of the collected data to extrapolate the flux maintenance value to the time point where the flux output decreases to a predetermined level (for example, 70% of initial flux). This time point is the flux maintenance life. The same curve-fit of the collected data can also be used to determine the flux output level at specific time points (e.g., 25,000 hours, 35,000 hours).

* Per ANSI/IES LM-80; see **Section 2.0 Normative References**.

This method is applied separately for each set of DUT test data collected at each operational (e.g., drive current) and environmental (e.g., case temperature) condition, as specified in ANSI/IES LM-80.

The procedures below outline the process for luminous flux maintenance projection. The same mathematical steps shall be followed for photon flux maintenance projection or radiant flux maintenance projection, with the substitution of photon flux data and use of Q_p or radiant flux data and use of R_p , respectively.

5.2 Procedures

5.2.1 Normalization. All collected data shall be normalized to a value of 1 (100%) at 0 hours for each DUT tested.

5.2.2 Average. The normalized measured data of all samples within the same data set defined in **Section 5.1** for each test condition at each measurement point shall be averaged.

5.2.3 Data Used for Curve-Fit. Data shall be collected according to the method described in **Section 4.3**.

For data sets of test duration D from 6,000 hours up to 10,000 hours, the data used for the curve-fits shall be applied to the last 5,000 hours of data. Data before the 1,000-hour reading shall not be used for the curve-fit.

The data collection points used in the projection calculation shall be at regular intervals (to within 48 hours). No two consecutive data collection intervals after the initial 1,000 hours shall differ by more than 96 hours in length. For example, data may be used in the projection calculation which are collected every 1,000 hours (± 48 hours), or every 500 hours (± 48 hours), but not every 1,000 hours and occasionally at 500 hours, since this would give excessive statistical weight to certain data points.

For data sets of test duration greater than 10,000 hours, the data for the last 50% of the total test duration shall be used for the curve-fit. In other words, all data points between $D/2$ and D shall be used. For example, if the test duration is 13,000 hours, all data points between

6,500 hours and 13,000 hours are used. If there is no data point at $D/2$, then data from the next lower time point shall be included in the curve fit. For example, if the test duration is 13,000 hours, with data taken every 1,000 hours, all data points between 6,000 and 13,000 hours are used.

5.2.4 Curve-Fit. An exponential least-squares curve-fit is applied to the averaged values, as specified in **Section 5.2.3**, for the following equation (see **Annex E** for calculation examples):

$$\Phi(t) = B e^{-\alpha t},$$

where:

t = operating time in hours

$\Phi(t)$ = averaged normalized flux output at time t

B = projected initial constant derived by the least-squares curve-fit

α = decay rate constant derived by the least-squares curve-fit

Use the following equations to project the luminous flux maintenance life:

$$\tau_{70} = \ln(B/0.7)/\alpha$$

or:

$$\tau_{90} = \ln(B/0.9)/\alpha$$

For any level of luminous flux maintenance, use the following generic form of the equation:

$$\tau_p = \ln(100B/p)/\alpha,$$

where:

τ_p = flux maintenance life expressed in hours, where $\tau = L$ for luminous flux, Q for photon flux, or R for radiant flux, and p is the percentage of initial flux output that is maintained.

When $\alpha < 2.0 \times 10^{-6}$, then the value of α shall be set to 2.0×10^{-6} . To avoid step changes between collected data and projected flux maintenance, when the minimum value of α is used the value of B shall be recalculated based on the last LM-80 data interval.

$$B_{\min\alpha} = \Phi_f e^{\alpha t},$$

where:

$B_{\min\alpha}$ = B value when minimum α value is used

Φ_f = flux value at last LM-80 measurement interval

t = last LM-80 measurement interval

Whenever a Φ (normalized flux) value is reached experimentally in the course of ANSI/IES LM-80 testing, the reported value shall be obtained by linear interpolation between the two nearest test points and takes precedence over any value projected by the formulas above.

5.2.5 Adjustment of Results. For a sample size of 20 units or more, flux maintenance values shall not be projected beyond 6 times the total test duration (in hours) of measured data. For a sample size of 10 to 19 units, flux maintenance values shall not be projected beyond 5.5 times the total test duration of measured data.

When the calculated flux maintenance life (e.g., τ_{70}) is less than or equal to 6 times (5.5 times for a sample size of 10 to 19 units) the total test duration, the calculated flux maintenance life is the reported flux maintenance life.

When the calculated flux maintenance life (e.g., τ_{70}) is greater than 6 times (5.5 times for a sample size of 10 to 19 units) the total test duration, the reported flux maintenance life value is limited to 6 times (5.5 times for a sample size of 10 to 19 units) the total test duration.

5.2.6 Notation for Projected Flux Maintenance Life.

The flux maintenance life projected in this method shall be expressed using the following notation:

$\tau_p (D \text{ k}) [y, z]$,

where:

τ = L for luminous flux, Q for photon flux, or R for radiant flux and p is the percentage of initial flux output that is maintained

D = the total duration of the test in hours divided by 1,000 and rounded to a nearest integer.

y = the minimum wavelength evaluated in nm (for photon flux or radiant flux)

z = the maximum wavelength evaluated in nm (for photon flux or radiant flux)

For example:

$L_{70}(6k)$ indicates 6,000 hours of test data for luminous flux maintenance

$Q_{90}(10k) [400, 700]$ indicates data from 400 to 700 nm and 10,000 hours of test data for photon flux maintenance

$R_{70}(8k) [350, 850]$ indicates data from 350 to 850 nm and 8,000 hours of test data for radiant flux maintenance

If the calculated τ_p value is limited to the test duration (see **Section 5.2.5**), the flux maintenance life value shall be expressed with a "greater-than" (>) symbol. For example:

- $\tau_{70}(6k) > 36,000$ hours (at $\tau_s = 55^\circ\text{C}$, $I_F = 350$ mA) for a sample size of 20 or more units
- $\tau_{70}(6k) > 33,000$ hours (at $\tau_s = 55^\circ\text{C}$, $I_F = 350$ mA) for a sample size of 10 to 19 units

If the τ_p value is reached experimentally in the course of ANSI/IES LM-80 testing, then the flux maintenance life shall be expressed with the D value equal to the τ_p value in hours divided by 1,000 and rounded to a nearest integer. For example:

$\tau_{70}(4k) = 4,400$ hours (at $\tau_s = 55^\circ\text{C}$, $I_F = 350$ mA)

5.2.7 Limit for Projecting Flux Maintenance.

As explained in **Annex A** it is possible for the flux maintenance behavior of a DUT to change over time and at different flux maintenance levels. For this reason, this method shall not be used to project lumen, photon, or radiant flux maintenance below 70%.

6.0 Temperature Data Interpolation

When in-situ DUT case temperature, $T_{s,ir}$ is different from the temperatures used for ANSI/IES LM-80 tests (e.g., 55°C , 85°C , and a third temperature provided by the DUT manufacturer), the following procedures shall be used to predict flux maintenance life of the DUT

corresponding to the in-situ case temperature with the same operational condition (e.g., drive current).

6.1 Select Tested Case Temperatures

The tested case temperatures used for in-situ case temperature flux maintenance life interpolation shall include the closest lower temperature, $T_{s,1}$, and the closest higher temperature, $T_{s,2}$, to the in-situ case temperature to be interpolated.

6.2 Convert All Temperatures to Kelvins

The following formula shall be used to convert the temperatures to the unit of kelvins:

$$T_s[\text{K}] = T_s[^\circ\text{C}] + 273.15$$

Only values expressed in kelvins shall be used in the calculations shown in the following sections.

6.3 Use the Arrhenius Equation to Calculate the Interpolated Flux Maintenance Life

The Arrhenius equation shown below shall be used to calculate in-situ decay rate constant α_i :

$$\alpha_i = A \exp\left(\frac{-E_a}{k_B T_{s,1}}\right),$$

where:

A = pre-exponential factor

E_a = activation energy (eV)

$T_{s,i}$ = in-situ absolute temperature (K)

k_B = Boltzmann's constant (8.617385×10^{-5} eV/K)

The following intermediate calculation steps need to be carried out in order to find the decay rate constant α_i at an in-situ temperature $T_{s,i}$ between $T_{s,1}$ and $T_{s,2}$.

Step 1: Obtain α_1 and α_2 for the two temperatures $T_{s,1}$ and $T_{s,2}$, which were calculated in the curve-fits performed per **Section 5.2.4**. Calculate the ratio of E_a/k_B :

$$E_a/k_B = (\ln \alpha_1 - \ln \alpha_2) / \left(\frac{1}{T_{s,2}} - \frac{1}{T_{s,1}} \right)$$

Step 2: Using the above ratio, plug in $T_{s,1}$ to calculate pre-exponential factor A :

$$A = \alpha_1 \exp\left(\frac{E_a}{k_B T_{s,1}}\right)$$

The above equation can also be used for substituting α_1 and $T_{s,1}$ with α_2 and $T_{s,2}$.

Step 3: Calculate B_0 :

$$B_0 = m_B T_{s,i} + b_B,$$

where:

$T_{s,i}$ = interpolation temperature

$m_B = (B_2 - B_1) / (T_{s,2} - T_{s,1})$

$b_B = B_1 - m_B T_{s,1}$

B_1 = projected initial constant for lower temperature ($T_{s,1}$) case

B_2 = projected initial constant for higher temperature ($T_{s,2}$) case

Step 4: Using the above result for B_0 , calculate flux maintenance life, τ_p , for in-situ case temperature $T_{s,i}$:

$$\tau_p = \ln(100 B_0 / p) / \alpha_i$$

Results from the above calculation steps, as well as the parameters used in the calculations, are reported in **Table 10-2** (see **Section 10.0 Report**).

Step 5: Calculate the in-situ flux output at time t , $\Phi_i(t)$, for $T_{s,i}$ as:

$$\Phi_i(t) = B_0 \exp(-\alpha_i t)$$

The reader is referred to **Annex E** for calculation examples.

7.0 Current Data Interpolation

When in-situ DUT drive current, $I_{s,i}$, is different from the drive current used for ANSI/IES LM-80 tests (e.g., ANSI/IES LM-80 DUT drive current = 350 mA and 700 mA, in-situ current = 520 mA), the following procedures shall be used to predict flux maintenance life of the DUT corresponding to the drive current with the same case temperature.

7.1 Select Tested Drive Currents

The tested drive currents used for drive current luminous flux maintenance life interpolation shall contain the

closest lower drive current, $I_{s,1}$, and the closest higher drive current, $I_{s,2}$, to the drive current to be interpolated. For DUTs with a maximum input power of 0.6 W or lower, the difference in drive current between any two data points being used for current interpolation shall not exceed a factor of 1.5. For DUTs with a maximum input power greater than 0.6 W, the difference in drive current between any two data points being used for current interpolation shall not exceed a factor of 2.

7.2 Interpolate Between Drive Currents

Use a linear function of temperature to interpolate α , used to calculate the interpolated flux maintenance life.

Step 1: The linear equation below shall be used to calculate in-situ decay rate constant α_i :

$$\alpha_i = m_\alpha I_{s,i} + b_\alpha,$$

where:

$$m_\alpha = (\alpha_2 - \alpha_1)/(I_{s,2} - I_{s,1})$$

$$b_\alpha = \alpha_1 - m_\alpha I_{s,1}$$

Step 2: Calculate B_0 :

$$B_0 = m_B I_{s,i} + b_B,$$

where:

$$m_B = (B_2 - B_1)/(I_{s,2} - I_{s,1})$$

$$b_B = B_1 - m_B I_{s,1}$$

B_1 = projected initial constant for lower temperature ($T_{s,1}$) case

B_2 = projected initial constant for higher temperature ($T_{s,1}$) case

Step 3: Using the above result of B_0 , calculate the flux maintenance life, or τ_p , for drive current $I_{s,i}$ as:

$$\tau_p = \ln(100 B_0 / p) / \alpha_i$$

Results from the above calculation steps, as well as the parameters used in the calculations, are reported in the **Table 10-2** (see **Section 10.0 Report**).

Step 4: Calculate the in-situ flux output at time t , $\Phi_i(t)$, for $I_{s,i}$ as:

$$\Phi_i(t) = B_0 \exp(-\alpha_i t)$$

The reader is referred to **Annex E** for calculation examples.

8.0 Simultaneous Temperature and Current Interpolation

When in-situ DUT drive current, $I_{s,i}$, and in-situ DUT case temperature, $T_{s,i}$, are different from the ANSI/IES LM-80 values, the following procedures shall be used to predict luminous flux maintenance life of the DUT corresponding to the drive current and in-situ DUT case temperature.

In general, this procedure requires a total of four data points. For clarity, the low temperature, low current point is referred to using the subscript (1, 1). The low temperature, high current data point is referred to using the subscript (1, 2). Similarly, the high temperature, low current data point is referred to using the subscript (2, 1). The high temperature, high current data point is referred to using the subscript (2, 2).

As with current interpolation, the difference in drive current between any two data points being used for current interpolation for DUTs with a maximum rated input power of 0.6 W or less shall not exceed a factor of 1.5. The difference in drive current between any two data points used for current interpolation for DUTs with a maximum rated input power greater than 0.6 W shall not exceed factor of 2.0.

8.1 Current Data Interpolation for the Lower In-Situ DUT Case Temperature Conditions

This is the in-situ DUT case temperature at the closest temperature below the interpolation temperature, T_1 , that includes at least two drive currents such that at least one drive current is above the interpolation drive current and one drive current is below the drive current. Current interpolation at the highest temperature that meets these requirements shall be used to create a first-temperature interpolation point using the current interpolation procedure as per **Section 7.1**. The closest drive currents to the interpolation current shall be used (i.e., the closest drive current below the interpolation drive current, $I_{1,1}$, and the closest drive current above the interpolation current, $I_{1,2}$. If a data point exists at the same drive current as the interpolation current and is at the closest lower temperature to the interpolation temperature, that data point shall be used for interpolating with the higher temperature condition.

When this type of interpolation is necessary, the following formulas shall be used:

$$\begin{aligned} m_{\alpha,1} &= (\alpha_{1,2} - \alpha_{1,1}) / (I_{1,2} - I_{1,1}) \\ b_{\alpha,1} &= \alpha_{1,2} - m_{\alpha,1} I_{1,2} \\ \alpha_{i,1} &= m_{\alpha,1} I_i + b_{\alpha,1} \\ m_{B,1} &= (B_{1,2} - B_{1,1}) / (I_{1,2} - I_{1,1}) \\ b_{B,1} &= B_{1,2} - m_{B,1} I_{1,2} \\ B_{i,1} &= m_{B,1} I_i + b_{B,1} \end{aligned}$$

8.2 Current Data Interpolation for the Higher In-Situ DUT Case Temperature Conditions

This is the in-situ DUT case temperature at the closest temperature above the interpolation temperature, T_2 , that includes at least two drive currents such that at least one drive current is above the interpolation drive current and one drive current is below the drive current. Current interpolation at the lowest temperature that meets these requirements shall be used to create a second temperature interpolation point using the current interpolation procedure as per **Section 7.1**. The closest drive currents to the interpolation shall be used (i.e., the closest drive current below the interpolation drive current, $I_{2,1}$, and the closest drive current above the interpolation current, $I_{2,2}$. If a data point exists at the same drive current as the interpolation current and is at the closest higher temperature to the interpolation temperature, that data point shall be used for interpolating with the lower temperature condition.

When this type of interpolation is necessary, the following formulas shall be used:

$$\begin{aligned} m_{\alpha,2} &= (\alpha_{2,2} - \alpha_{2,1}) / (I_{2,2} - I_{2,1}) \\ b_{\alpha,2} &= \alpha_{2,2} - m_{\alpha,2} I_{2,2} \\ \alpha_{i,2} &= m_{\alpha,2} I_i + b_{\alpha,2} \\ m_{B,2} &= (B_{2,2} - B_{2,1}) / (I_{2,2} - I_{2,1}) \\ b_{B,2} &= B_{2,2} - m_{B,2} I_{2,2} \\ B_{i,2} &= m_{B,2} I_i + b_{B,2} \end{aligned}$$

8.3 Temperature Interpolation Between Lower and Higher Current Interpolated In-Situ Case Temperatures

Temperature interpolation is performed as per **Section 6.3** using the interpolated current values at the lower in-situ DUT case temperature, T_1 , and the higher in-situ DUT case temperature, T_2 .

Step 1: Convert in-situ DUT case temperatures to kelvins:

$$\begin{aligned} T_{A,1}[\text{K}] &= T_1[^\circ\text{C}] + 273.15 \\ T_{A,2}[\text{K}] &= T_2[^\circ\text{C}] + 273.15 \\ T_{A,i}[\text{K}] &= T_i[^\circ\text{C}] + 273.15 \end{aligned}$$

Step 2: Calculate the ratio of E_a/k_B :

$$\frac{E_a}{k_B} = \frac{\ln(\alpha_{i,1}) - \ln(\alpha_{i,2})}{\frac{1}{T_{A,2}} - \frac{1}{T_{A,1}}}$$

Step 3: Calculate the pre-exponential factor A :

$$A = \alpha_{i,1} \exp\left(\frac{E_a}{k_B T_{A,1}}\right)$$

Step 4: Calculate the in-situ DUT case temperature interpolated value of α :

$$\alpha_i = A \exp\left(\frac{-E_a}{k_B T_{A,i}}\right)$$

Step 5: Calculate B_0 :

$$m_B = (B_{i,2} - B_{i,1}) / (T_{A,2} - T_{A,1}),$$

where:

$$\begin{aligned} b_B &= B_{i,1} - m_B T_{A,1} \\ B_0 &= m_B T_{A,i} + b_B \end{aligned}$$

Step 6: Calculate flux maintenance life, or τ_p , for the interpolated drive current and interpolated in-situ DUT case temperature:

$$\tau_p = \frac{\ln\left(\frac{100 B_0}{p}\right)}{\alpha_i}$$

Results from the above calculation steps, as well as the parameters used in the calculations, are reported in **Table 10-2** (see **Section 10.0 Report**).

The reader is referred to **Annex E** for calculation examples.

9.0 Limits for Extrapolation

Extrapolation of any τ_p , (luminous flux maintenance life, photon flux maintenance life, or radiant flux

maintenance life) value above the operating temperature used in the ANSI/IES LM-80 test shall not be performed. For example, if the highest ANSI/IES LM-80 test temperature is 105 °C, a τ_p at 110°C shall not be extrapolated.

If the actual operating temperature is below the lowest temperature that is used in the ANSI/IES LM-80 test, the reported τ_p value shall be that using the lowest temperature in the ANSI/IES LM-80 test. For example, if the lowest ANSI/IES LM-80 temperature is 55 °C, the report for 45 °C shall use the τ_p at 55°C.

Extrapolation of any τ_p value above the drive current used in the ANSI/IES LM-80 test shall not be performed. For example, if the highest ANSI/IES LM-80 drive current is 700 mA, a τ_p at 750 mA shall not be extrapolated.

If the actual drive current is below the lowest drive current that is used in the ANSI/IES LM-80 test, the reported τ_p value shall be that using the lowest drive current in the ANSI/IES LM-80 test. For example, if the lowest ANSI/IES LM-80 drive current is 80 mA, the report for 50 mA shall use the τ_p at 80 mA.

For interpolation between case temperatures or drive currents with test durations of unequal length, projections shall not extend beyond 6 times the shortest

test duration for 20 or more samples (5.5 times the shortest test duration for 10 to 19 samples).

10.0 Report

The report of flux maintenance life projection shall include the following information included as shown in the recommended format of **Table 10-1**. The reported τ_{70} values shall be rounded to 3 significant digits. The values for α and B shall be rounded to 4 significant digits.

When temperature interpolation is used, additional information shall be presented, as shown in the recommended format of **Table 10-2** (refer to **Section 6** for definitions).

When current interpolation is used, additional information shall be presented, as shown in the recommended format of **Table 10-3** (refer to **Section 7** for definitions).

When current and temperature interpolation is used, the additional following information shall be presented, as shown in the recommended format of **Table 10-4** (refer to **Section 8** for definitions).

Table 10-1. Recommended Report Information for Each ANSI/IES LM-80 Test Condition

Description of LED light source tested (e.g., manufacturer, model, catalog number)	
Sample size	
DUT drive current used in the test	_____ mA
Test duration	_____ hours
Test duration used for projection	_____ hour to _____ hour
Tested case temperature	_____ °C
Flux maintenance	
α	
B	
Reported $\tau_{70}(Dk)$ [y, z]*	_____ hours

* The appropriate flux maintenance parameter shall replace τ in the report (L_{70} for luminous flux maintenance life, Q_{70} for photon flux maintenance life, R_{70} for radiant flux maintenance life).

Table 10-2. Recommended Report Information for Temperature Interpolation

$T_{s,2}$ (K)	
α_2	
B_2	
$T_{s,i}$ (°C)	
$T_{s,i}$ (K)	
α_i	
B_0	
Reported τ_{70} (D k) [y, z]*	

Table 10-3. Recommended Report Information for Current Interpolation

α_2	
B_2	
$I_{s,i}$ (mA)	
α_i	
B_0	
Reported τ_{70} (D k) [y, z]*	

* The appropriate flux maintenance parameter shall replace τ in the report (L_{70} for luminous flux maintenance life, Q_{70} for photon flux maintenance life, R_{70} for radiant flux maintenance life).

Table 10-4. Recommended Report Information for Current and Temperature Interpolation

$T_{s,1}$ (°C)			$T_{s,2}$ (°C)			$I_{s,i}$ (mA)	
$T_{A,1}$ (K)			$T_{A,2}$ (K)			$T_{s,i}$ (°C)	
$I_{1,1}$ (mA)			$I_{2,1}$ (mA)			$T_{A,i}$ (K)	
$B_{1,2}$			$B_{2,2}$			α_i	
$\alpha_{i,1}$			$\alpha_{i,2}$				
$B_{i,1}$			$B_{i,2}$				

The calculated and projected $\tau_p(Dk)$ values are not to be reported in **Tables 10-1** through **10-4** if they exceed the allowable extrapolation limits of **Section 5.2.5**. These values are set for the purpose of calibrating the calculation tools for the users. Due to the statistical

uncertainty stated in **Annex D**, any calculated result in hours beyond the reported projection limit set in **Section 9** does not have any practical or meaningful value.

Annex A – Rationale for the Recommended Data Extrapolation Method

The original IES TM-21 Working Group (WG) conducted statistical analyses for over 40 sets of IES LM-80-08 test data collected from four major LED manufacturers, in which more than 20 data sets had testing duration greater than or equal to 10,000 hours. Several mathematical models were proposed to predict luminous flux maintenance life based on these real LED luminous flux maintenance data, which exhibited several different trends. The WG conducted thorough investigations in several areas.

First, several possible models that express potential LED physical flux-decay behaviors were identified and proposed and a search made for possible metrics that can be used to evaluate and select the most accurate model for a given set of IES LM-80-08 data assuming that the duration is 6,000 hours. Second, the WG evaluated the LED flux maintenance behaviors that are shown in the 40-plus sets of data and noticed that many LEDs exhibit some rapid changes over the first 1,000 or more hours. The WG studied various options, including using data only after 1,000 hours or only after the first hump of the flux maintenance curve. This was to verify whether a chosen mathematical model can be used more reliably. Third, the WG examined the accuracy of prediction using various proposed models by examining the IES LM-80-08 data that extends to longer hours, e.g., 10,000 to 15,000 hours. These explorations showed that the statistics of model fitting using only 6,000 hours of IES LM-80-08 data were not conclusive enough to realistically help in identifying the most suitable model to represent lumen output degradation. The explorations also showed that the LED flux depreciation trends often change after 6,000 hours in one way or another, and there is no reliable and consistent approach to predict such trends from the 6,000-hour data points. Last, natural “noise” in the real data can falsely indicate decay trends in the 6,000-hour test, which test data longer than 6,000 hours show is not real. Some noisy data points are inevitable, due to measurement uncertainties of the luminous flux

measurements over a long period of time, and the best fit for the given 6,000-hour data points would not assure that prediction for much longer time points would be accurate.

The WG further discussed whether a possible better model (other than exponential) could be used for test data longer than 6,000 hours. Again, some real LED data showed changes in the lumen depreciation trend after 10,000 hours, and the same problems are observed. It is also understood by LED manufacturers that LED luminous flux maintenance curves tend to change depending on the technologies and materials used in the LED packages. Some of these occur at early times, and some later than 6,000 hours or even 10,000 hours. The analysis completed on the longer data sets confirmed that selection of a “better” mathematical model is not appropriate even with 10,000-hour data.

In many cases, the statistical-fit data for multiple models showed little difference, indicating that more than one model may be reasonable (given the uncertainty of long-reaching extrapolation) in determining the expected lumen depreciation of an LED. There is also collective understanding within the LED industry that degradation associated with components other than the diode itself tends to manifest itself earlier rather than later. Therefore, the more stable lumen depreciation in later time segments is mostly associated with diode degradation, and this has been considered reasonable, as it corresponds to classic exponential decay.

Therefore, the WG concluded that the most reasonable approach to extrapolation of lumen degradation when removing initial variable data (associated with a hump in the data) was the use of the simple exponential fit. With further analyses of the collected test data, the WG found that for longer test duration, e.g., 10,000 hours or longer, using the last 5,000 hours of the data (5,000 to 10,000 hours) showed more consistent and reliable prediction results than using the data for the entire test duration (1,000 to 10,000 hours). However, there was a concern that the last 5,000 hours provides only 6 data points (assuming the test interval is 1,000 hours), and the results are susceptible to noisy data points. After further discussions and analyses, the WG determined that the last 50% of the test data, with a minimum of

the last 5,000 hours' data, would be an appropriate application of the available data for extrapolations. The WG conducted verification extrapolations and found that this method works well without any serious problems for all the collected LED test data. It is noted that these test data used for verification are still limited to up to approximately 15,000 hours.

Some examples of real LED test data and lumen-maintenance curve fitting are shown in **Figure A-1**, which is an example where the exponential fit for 1,000 to 6,000 hours appears very good, but the trend changes after 6,000 hours. In this case, $L_{70}(6k)$ is 60,000 hours, but $L_{70}(10k)$ is 30,000 hours, which is considered more accurate. **Figure A-2** is another example where the 1,000- to 6,000-hour data fits very well in that portion, but the trend changes after 6,000 hours in the other direction. In this case, $L_{70}(6k)$ is 30,000 hours, but $L_{70}(10k)$ is 60,000 hours (limited by a value of 6 times the total test duration), which is considered to be more accurate. In both cases, it is not possible to predict the change in the trend only from the 6,000-hour data points.

As examples shown in the above **Figures A-1** and **A-2**, the L_{70} prediction taken from 6,000 hours of test data is often found unreliable and unsatisfactory. The longer the test duration, the more accurate the prediction will be. Thus, it is recommended by the WG that ANSI/IES

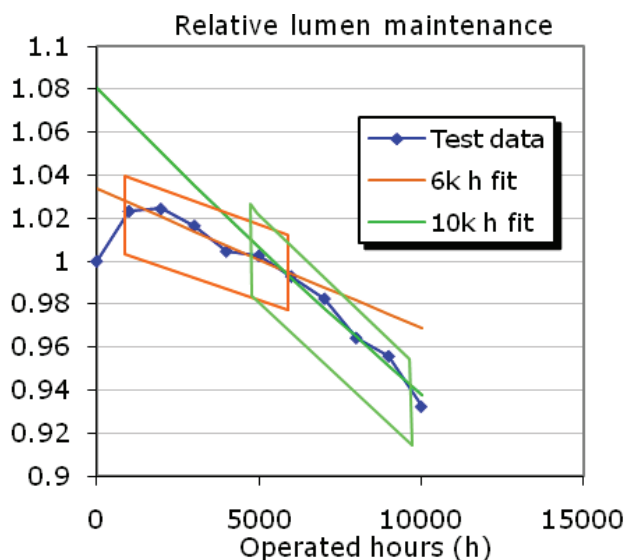


Figure A-1. LED data, first example.

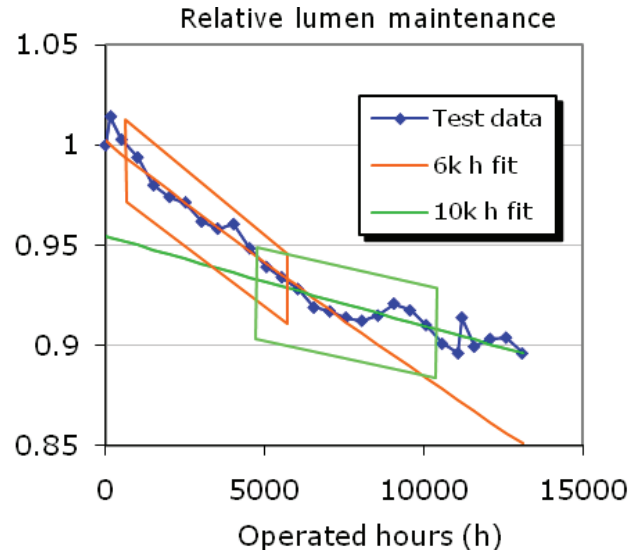


Figure A-2. LED data, second example.

LM-80 tests should be continued after 6,000 hours and that τ_{70} predictions should be updated when test data at longer durations becomes available. To encourage such practice and allow the users to know the length of test data from which τ_{70} is determined, the notation, $\tau_p(D \text{ k})$ (see **Section 5.2.6**), is introduced so that the duration of the test is always reported with the life value. The duration of the test hours helps indicate the degree of reliability of the reported luminous flux maintenance life.

Because LEDs often show an initial flux maintenance behavior that differs from the long-term flux maintenance behavior, the WG instituted a moving time window as a way to encourage collection of data points after the initial 6,000 hours of ANSI/IES LM-80 testing. For this reason, the WG also chose to use an extrapolation limit based on the test duration rather than on the time interval used for establishing the projection.

These rules mean that the data from 1,000 hours to 6,000 hours may be used to extrapolate to 36,000 hours or 30,000 hours past the last test interval. If the test data extended to 8,000 hours, then the data from 3,000 hours to 8,000 hours can be extrapolated to 48,000 hours or 40,000 hours past the last test interval, despite being based on the same interval of time.

Annex B – Sample Size Selection for Data Extrapolation

As explained in **Annex A**, the exponentially modeled mean values are used to establish the LED light source decay structure. This fitted exponential model is then used to extrapolate the expected flux output. When using the simplified structure of fitting to the mean at each time point, it is important to identify how many units are tested. Increasing the number of units tested at each time point will provide a stronger foundation for the fitted exponential model. **Figure B-1** shows the relative uncertainty in calculated flux maintenance life (y-axis) at each stepped increase in the number of units used to calculate the flux maintenance life (the sample size). As can be seen, the uncertainty in flux maintenance life decreases as the number of units increases.

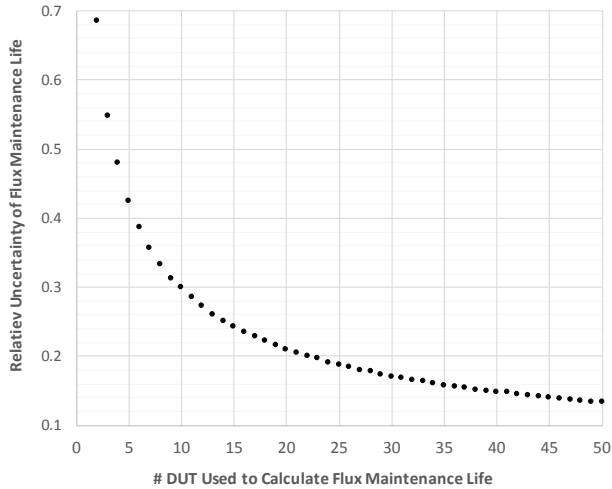


Figure B-1. Sample size versus uncertainty.

Another concern in model fitting rests on the assumption of the normality of the uncertainty error associated with the fitted model. In this application a similar evaluation of measuring a sufficient number of units at each time point can provide a substantial basis for the normality of the means. Most LED test data collected by the TM-21 Working Group (WG) have an approximately symmetrical distribution about the mean of the data. This symmetry and, consequently, limited skewness

provide justification that the normality assumptions can be reasonably accepted with as few as 20 units per time period.

Therefore, the WG made the recommendation for sample size requirement found in **Section 4.2**.

Annex C – Least-Squares Formula

For a set of n experimental data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the least-squares straight line fit, where n is the total number of averaged data points used for curve-fitting, according to **Section 5.2.4**.

$$y = mx + b,$$

where m is the slope:

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

and b is an intercept:

$$b = \frac{\sum y - m \sum x}{n}$$

and Σ stands for a sum of n terms:

$$\sum xy = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$$\sum x = x_1 + x_2 + \dots + x_n$$

$$\sum y = y_1 + y_2 + \dots + y_n$$

$$\sum x^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

After the substitutions of:

$$x_k = t_k, \text{ where } k = 1, 2, \dots, n$$

and

$$y_k = \ln \Phi_k, \text{ where } k = 1, 2, \dots, n$$

Then, the above formula can be used to derive B and a :

$$B = e^b$$

and

$$\alpha = -m$$

Annex D – Limit for Duration of Prediction

This annex describes the analysis used to determine a multiplication factor for the upper limit of τ_{70} that is to be reported. This analysis focused on τ_{70} .

The original IES TM-21 Working Group (WG) conducted analyses on over 40 sets of IES LM-80-08 test data collected from several LED manufacturers. To determine the uncertainty of the exponential curve-fit to the datasets, a confidence band was calculated, which shows the region within which the model is expected to fall with a certain level of probability. A confidence band is calculated using the Student's t function, the coefficients of the model, and the estimated uncertainties in the coefficients; therefore, a covariance matrix of the predictive model is required. Thus, to calculate a confidence band an estimation of uncertainty is required for each data point. An example of a calculated confidence band is presented in **Figure D-1**.

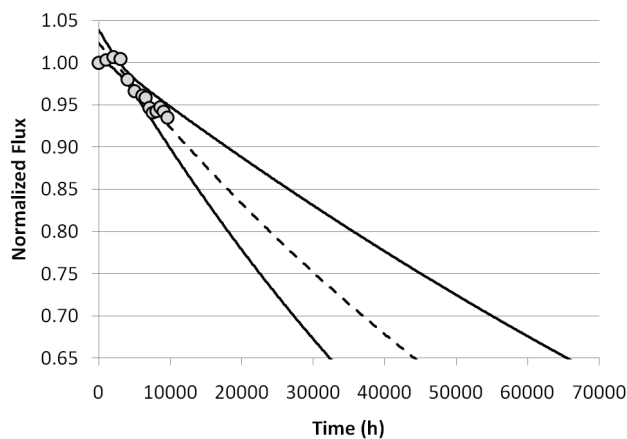


Figure D-1. The grey circles are the normalized data. The dashed line is the optimized curve-fit to the data for 1,000 to 6,000 hours. The solid lines are the confidence bands with respect to the optimized fit.

For this analysis, two main components were combined as the square root of a sum of squares. The first component is the standard deviation of an individual dataset for a given time, divided by the square root of the number of points, i.e., the standard deviation of the mean value. The second component is the uncertainty

of the measurement system for relative measurements over the time frame of the measurements (the test duration). This is a characterization of the reproducibility of the measurement system. The magnitude of the reproducibility is usually characterized by a control chart documented over the data collection time. A matrix of these two components was analyzed. The number of points used to determine the mean standard deviation was varied: 5, 10, 20, 30, 50, and 100 points. A bootstrap method was used to increase the number of data points for a given time. The relative combined uncertainty of the measurement system was also varied: 0.10%, 0.25%, 0.40%, 0.50%, 0.75% and 1.0%. The relative combined uncertainty of the measurement system is the uncertainty between measurements for the same device over the duration of the test. It does not include the absolute calibration of the measurement system. It is a measure of the system stability over time. The expanded uncertainty with a coverage factor of $k=2$, which would represent a 95% coverage interval, is 0.2%, 0.5%, 0.8%, 1.0%, 1.5%, and 2.0%. The level of probability for this analysis was set at 90%, using a one-sided distribution for the lower confidence band; therefore, the calculated τ_{70} is 9 times out of 10 going to be greater than the cutoff determined by the lower confidence band.

In order to compare the more than 40 data sets at one time, a test statistic was created. To determine a multiplication factor for the upper limit of τ_{70} , to be reported with a confidence of 90%, a hypothesis was created that if the statistic was greater than 1, the hypothesis is considered true. To calculate the statistic, a multiplier to be tested is chosen, for example, 6. For a given set of data, the τ_{70} is calculated and the lower confidence band is calculated. The critical time for a multiplier of 6 is the time interval (6,000 hours) times the multiplier plus 1, or 42,000 hours. If the lower confidence band is greater than the cut-off of 36,000 hours, the hypothesis is true.

To plot this statistic, the lower confidence band (LCB) is divided by the τ_{70} value, multiplied by the multiplier (M) plus 1, and divided by the multiplier.

$$\text{Test statistic} = \frac{LCB(M+1)}{\tau_{70}M}$$

Simply put, is the difference between the lower confidence band and τ_{70} the same magnitude as the data collection total time?

Although errors in the decay rate constant (α) are symmetric, Φ_i is an exponential function of α which means the errors in Φ_i are asymmetric. This can be observed by inspection of the solid lines in **Figure D-1**. The uncertainty limits correspond to -14.3%, +20.0% or a mean uncertainty of 16.9%.

Stated another way, the 80% confidence interval for τ_{70} = 30,000 hours would be 25,700 to 36,000 hours.

Figure D-2 shows that the statistic has a value of 1 at the test time of 36,000 hours, within the uncertainty of the fit. The test statistic decreases with time because at 6,000 hours and an L_{70} of 200,000 hours, the data points provide very little curvature to fit the exponential. The conclusions of the analysis are that for a 0.40% relative combined uncertainty for the measurement system, which is typical for the industry at this time based on control charts provided by testing laboratories, a 6-times multiplier is statistically acceptable for data sets with at least 20 data points (sample units). A 5.5-times multiplier is statistically acceptable for data sets with 10 to 19 data points (sample units).

Annex E – Data Test Set for Validation of Calculation – Examples

To assist users of this document to carry out the calculations following the procedures listed in **Section 5** and **Section 6**, the examples of the calculations based on the ANSI/IES LM-80-15 data are presented in **Annex F**. The data used in the calculations were selected from the database submitted from LED manufacturers to the IES TM-21 Working Group (WG). It is recommended that users use the data and calculation results presented in this annex for comparison with their calculations to ensure the calculation steps programmed by the users are correct.

E.1 Examples of Normalizing and Fitting 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Data

Table E-1 represents the photon flux data sets from 20 units of tested samples for 6,000 hours of ANSI/IES LM-80-15 testing at case temperature 55 °C. **Table E-2** represents 6,000 hours of ANSI/IES LM-80-15 testing at case temperature 85 °C.

The results of the least-squares fit using this dataset is shown in **Table E-3** for case temperature of 55 °C and in **Table E-4** for 85 °C. It should be noted that data from 1,000 hours to 6,000 hours are used for calculation of projected luminous flux maintenance life.

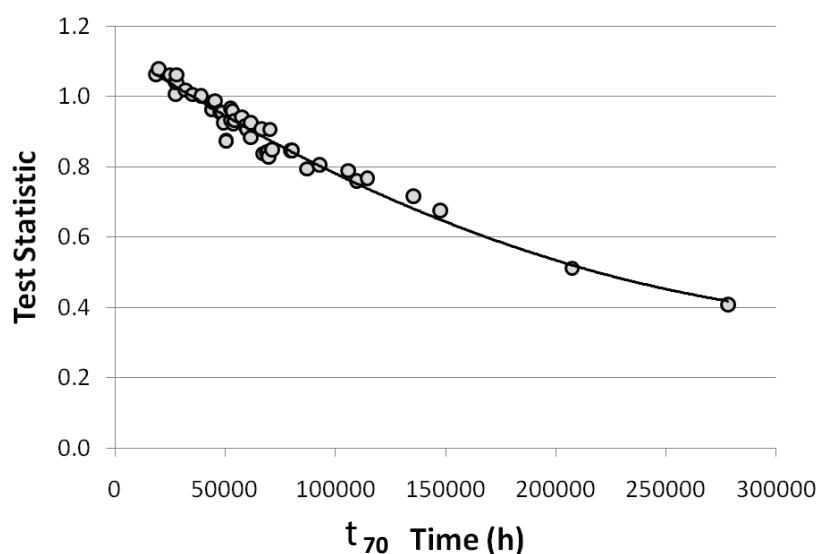


Figure D-2. The test statistic plotted versus the τ_{70} calculated for each dataset using 0.40% relative combined uncertainty for the measurement system and 20 data points. A quadratic curve was fit to the test statistic, which is based on a 6-times multiplier.

Table E-1. 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Test Data at Case Temperature Point $T_{s,1} = 55\text{ }^{\circ}\text{C}$ Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Sample #	Time (h)							
	0	500	1,000	2,000	3,000	4,000	5,000	6,000
1	1.000	0.970	0.957	0.962	0.957	0.950	0.944	0.947
2	1.000	0.987	0.973	0.976	0.971	0.967	0.960	0.960
3	1.000	0.984	0.966	0.967	0.960	0.954	0.947	0.949
4	1.000	0.990	0.977	0.980	0.976	0.970	0.967	0.965
5	1.000	0.981	0.963	0.969	0.965	0.959	0.953	0.953
6	1.000	0.988	0.975	0.979	0.974	0.968	0.964	0.966
7	1.000	0.990	0.978	0.978	0.974	0.962	0.958	0.954
8	1.000	0.988	0.973	0.974	0.968	0.962	0.957	0.955
9	1.000	0.989	0.975	0.978	0.974	0.968	0.964	0.966
10	1.000	0.982	0.965	0.964	0.957	0.948	0.942	0.936
11	1.000	0.977	0.956	0.960	0.956	0.950	0.946	0.946
12	1.000	0.988	0.975	0.980	0.977	0.970	0.967	0.961
13	1.000	0.985	0.969	0.971	0.965	0.956	0.949	0.945
14	1.000	0.976	0.960	0.966	0.962	0.957	0.953	0.953
15	1.000	0.985	0.971	0.978	0.975	0.969	0.965	0.966
16	1.000	0.977	0.962	0.969	0.964	0.958	0.956	0.952
17	1.000	0.966	0.950	0.954	0.944	0.938	0.935	0.937
18	1.000	0.998	0.983	0.989	0.984	0.977	0.972	0.971
19	1.000	0.985	0.970	0.976	0.969	0.963	0.958	0.957
20	1.000	0.975	0.961	0.967	0.961	0.952	0.948	0.944
Average	1.0000	0.9831	0.9680	0.9719	0.9667	0.9599	0.9553	0.9542
ln(Average)	0.0000	-0.01704	-0.03252	-0.02850	-0.03387	-0.04093	-0.04573	-0.04688

Table E-2. 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Test Data at Case Temperature Point $T_{s,2} = 85^{\circ}\text{C}$ Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Sample #	Time (h)							
	0	500	1,000	2,000	3,000	4,000	5,000	6,000
1	1.000	0.995	0.969	0.972	0.957	0.944	0.933	0.929
2	1.000	0.986	0.961	0.968	0.958	0.946	0.938	0.937
3	1.000	0.969	0.951	0.951	0.938	0.923	0.918	0.917
4	1.000	0.988	0.972	0.973	0.959	0.95	0.948	0.947
5	1.000	0.971	0.95	0.95	0.936	0.922	0.911	0.907
6	1.000	0.974	0.956	0.953	0.941	0.927	0.919	0.914
7	1.000	0.988	0.971	0.974	0.966	0.956	0.95	0.95
8	1.000	0.985	0.969	0.976	0.965	0.956	0.951	0.95
9	1.000	0.986	0.967	0.969	0.954	0.938	0.93	0.924
10	1.000	0.949	0.922	0.921	0.907	0.894	0.885	0.885
11	1.000	0.993	0.978	0.982	0.974	0.966	0.961	0.959
12	1.000	0.991	0.976	0.977	0.97	0.959	0.953	0.949
13	1.000	0.981	0.963	0.972	0.966	0.956	0.95	0.952
14	1.000	0.992	0.976	0.982	0.972	0.962	0.958	0.958
15	1.000	0.967	0.947	0.943	0.932	0.92	0.914	0.914
16	1.000	0.984	0.967	0.973	0.965	0.941	0.94	0.94
17	1.000	0.992	0.977	0.982	0.971	0.962	0.956	0.957
18	1.000	0.984	0.967	0.967	0.952	0.939	0.932	0.928
19	1.000	0.981	0.964	0.964	0.953	0.939	0.933	0.929
20	1.000	0.982	0.966	0.97	0.96	0.951	0.948	0.941
Average	1.0000	0.9819	0.9635	0.9660	0.9548	0.9426	0.9364	0.9344
ln(Average)	0.0000	-0.01827	-0.03718	-0.03459	-0.04625	-0.05911	-0.06571	-0.06785

Table E3. Least-Squares Curve-Fit for 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Test Data at Temperature Point $T_{s,1}$ = 55 °C Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Point #	Time (h)	ln(Average)	x	y	xy	x ²
1	1,000	-0.03252	1000	-0.03252	-32.52	1.000E+06
2	2,000	-0.02850	2000	-0.02850	-57.00	4.000E+06
3	3,000	-0.03387	3000	-0.03387	-101.61	9.000E+06
4	4,000	-0.04093	4000	-0.04093	-163.72	1.600E+07
5	5,000	-0.04573	5000	-0.04573	-228.65	2.500E+07
6	6,000	-0.04688	6000	-0.04688	-281.28	3.600E+07
		Sums	21000	-0.2284	-864.78	9.100E+07
		Slope	-3.730E-06			
		Intercept	-0.02502			
		Reported α_1	3.730E-06			
		B_1	0.9753			
		Reported Q_{90}	21,500			

Table E4. Least-Squares Curve-Fit for 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Test Data at Temperature Point $T_{s,2}$ = 85 °C Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Point #	Time (h)	ln(Average)	x	y	xy	x ²
1	1,000	-0.03718	1000	-0.03718	-37.18	1.000E+06
2	2,000	-0.03459	2000	-0.03459	-69.18	4.000E+06
3	3,000	-0.04625	3000	-0.04625	-138.75	9.000E+06
4	4,000	-0.05911	4000	-0.05911	-236.44	1.600E+07
5	5,000	-0.06571	5000	-0.06571	-328.55	2.500E+07
6	6,000	-0.06785	6000	-0.06785	-407.10	3.600E+07
		Sums	21000	-0.3107	-1217.20	9.100E+07
		Slope	-7.416E-06			
		Intercept	-0.02582			
		Reported α_2	7.416E-06			
		B_2	0.9745			
		Reported Q_{90} [400,700]	10,700			

E.2 Example of Arrhenius Interpolation Using 6,000 Hours of ANSI/IES LM-80-15 Photon Flux Data

As example, if the in-situ case temperature is $T_{s,i} = 70^\circ\text{C}$, the data in **Table E-1** ($T_{s,1} = 55^\circ\text{C}$) and **Table E-2** ($T_{s,2} = 85^\circ\text{C}$) are used to interpolate, and to project the photon flux maintenance life for $T_{s,i} = 70^\circ\text{C}$. The parameters are shown as in **Table E-5**.

Table E-5. Parameters of Interpolation Using 6,000 Hours ANSI/IES LM-80-15 Photon Flux Data for In-Situ Case Temperature $T_{s,i} = 70^\circ\text{C}$ Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Parameter	Temperature		
	Condition 1	Condition 2	Interpolation
T_s ($^\circ\text{C}$)	55	85	70
T_s (K)	328.15	358.15	343.15
α	3.730E-06	7.416E-06	5.339E-06
B	0.9753	0.9745	0.9749
E_a/k_B	2692	N.A.	N.A
A	1.363E-02	N.A	N.A
m_B	-2.667E-05	N.A	N.A
b_B	0.9768	N.A	N.A
Reported Q_{90} [400, 700]	21,500	10,700	15,000

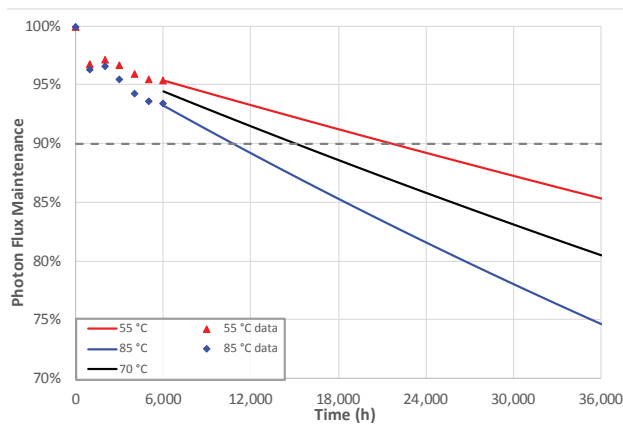


Figure E-1. Graphic representation of luminous flux maintenance life projection using 6,000 hours of ANSI/IES LM-80-15 data for temperature interpolation evaluated over the wavelength interval of 400 nm to 700 nm.

The graphic representation of the photon flux maintenance life projection results for $T_{s,1} = 55^\circ\text{C}$, $T_{s,2} = 85^\circ\text{C}$, and in-situ temperature of $T_{s,i} = 70^\circ\text{C}$ is shown in **Figure E-1**.

The numerical representation of the photon flux maintenance life projection results for $T_{s,1} = 55^\circ\text{C}$, $T_{s,2} = 85^\circ\text{C}$, and in-situ temperature of $T_{s,i} = 70^\circ\text{C}$ are shown in **Table E-6**.

Table E-6. Numerical Results of Photon Flux Maintenance Life Projection Using 6,000 Hours of ANSI/IES LM-80-15 Data Evaluated Over the Wavelength Interval of 400 nm to 700 nm

Time (h)	Photon Flux Maintenance [400, 700]		
	55 $^\circ\text{C}$	70 $^\circ\text{C}$	85 $^\circ\text{C}$
6,000	0.9537	0.9442	0.9321
7,000	0.9502	0.9391	0.9252
8,000	0.9466	0.9341	0.9184
9,000	0.9431	0.9292	0.9116
10,000	0.9396	0.9242	0.9048
11,000	0.9361	0.9193	0.8982
12,000	0.9326	0.9144	0.8915
13,000	0.9291	0.9095	0.8849
14,000	0.9257	0.9047	0.8784
15,000	0.9222	0.8999	0.8719
16,000	0.9188	0.8951	0.8655
17,000	0.9154	0.8903	0.8591
18,000	0.9120	0.8856	0.8527
19,000	0.9086	0.8809	0.8464
20,000	0.9052	0.8762	0.8402
21,000	0.9018	0.8715	0.8340
22,000	0.8985	0.8669	0.8278
23,000	0.8951	0.8622	0.8217
24,000	0.8918	0.8577	0.8156
25,000	0.8885	0.8531	0.8096
26,000	0.8852	0.8485	0.8036
27,000	0.8819	0.8440	0.7977
28,000	0.8786	0.8395	0.7918
29,000	0.8753	0.8351	0.7859
30,000	0.8720	0.8306	0.7801
31,000	0.8688	0.8262	0.7743
32,000	0.8656	0.8218	0.7686
33,000	0.8623	0.8174	0.7629
34,000	0.8591	0.8131	0.7573
35,000	0.8559	0.8087	0.7517
36,000	0.8527	0.8044	0.7462

E.3 Examples of Normalizing and Fitting 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Data

The examples shown in this section are for the radiant flux data collected for 10,000 hours of ANSI/IES LM-80-

15 testing. **Table E-7** represents the data sets from 20 units of tested samples for 10,000 hours of ANSI/IES LM-80-15 testing at case temperature 55 °C. **Table E-8** represents 10,000 hours of ANSI/IES LM-80-15 testing at case temperature 85 °C.

Table E-7. 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Test Data at Case Temperature Point $T_{s,1} = 55\text{ °C}$ Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Sample #	0	4,000	5,000	6,000	7,000	8,000	9,000	10,000
1	1.000	0.950	0.944	0.947	0.947	0.943	0.940	0.943
2	1.000	0.967	0.960	0.960	0.960	0.956	0.951	0.956
3	1.000	0.954	0.947	0.949	0.946	0.941	0.936	0.941
4	1.000	0.970	0.967	0.965	0.967	0.964	0.961	0.964
5	1.000	0.959	0.953	0.953	0.953	0.948	0.945	0.948
6	1.000	0.968	0.964	0.966	0.963	0.959	0.954	0.958
7	1.000	0.962	0.958	0.954	0.961	0.949	0.948	0.951
8	1.000	0.962	0.957	0.955	0.956	0.952	0.948	0.951
9	1.000	0.968	0.964	0.966	0.964	0.960	0.957	0.960
10	1.000	0.948	0.942	0.936	0.939	0.934	0.930	0.930
11	1.000	0.950	0.946	0.946	0.950	0.946	0.943	0.947
12	1.000	0.970	0.967	0.961	0.965	0.961	0.959	0.962
13	1.000	0.956	0.949	0.945	0.946	0.939	0.933	0.933
14	1.000	0.957	0.953	0.953	0.953	0.950	0.947	0.950
15	1.000	0.969	0.965	0.966	0.963	0.960	0.957	0.959
16	1.000	0.958	0.956	0.952	0.956	0.955	0.952	0.953
17	1.000	0.938	0.935	0.937	0.937	0.932	0.928	0.931
18	1.000	0.977	0.972	0.971	0.972	0.966	0.960	0.963
19	1.000	0.963	0.958	0.957	0.956	0.951	0.946	0.949
20	1.000	0.952	0.948	0.944	0.946	0.942	0.939	0.941
Average	1.0000	0.9599	0.9553	0.9542	0.9550	0.9504	0.9467	0.9495
ln(Average)	0.00000	-0.04093	-0.04573	-0.04688	-0.04604	-0.05087	-0.05477	-0.05182

Table E-8. 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Test Data at Case Temperature Point $T_{s,2} = 85\text{ }^{\circ}\text{C}$ Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Sample #	0	4,000	5,000	6,000	7,000	8,000	9,000	10,000
1	1.000	0.944	0.933	0.929	0.924	0.918	0.913	0.914
2	1.000	0.946	0.938	0.937	0.932	0.924	0.918	0.922
3	1.000	0.923	0.918	0.917	0.911	0.902	0.898	0.902
4	1.000	0.950	0.948	0.947	0.949	0.942	0.938	0.941
5	1.000	0.922	0.911	0.907	0.903	0.894	0.888	0.889
6	1.000	0.927	0.919	0.914	0.913	0.905	0.900	0.902
7	1.000	0.956	0.950	0.950	0.950	0.944	0.939	0.942
8	1.000	0.956	0.951	0.950	0.948	0.942	0.935	0.936
9	1.000	0.938	0.930	0.924	0.921	0.911	0.905	0.905
10	1.000	0.894	0.885	0.885	0.880	0.876	0.873	0.878
11	1.000	0.966	0.961	0.959	0.958	0.952	0.949	0.953
12	1.000	0.959	0.953	0.949	0.949	0.944	0.939	0.941
13	1.000	0.956	0.950	0.952	0.951	0.947	0.944	0.950
14	1.000	0.962	0.958	0.958	0.956	0.949	0.943	0.948
15	1.000	0.920	0.914	0.914	0.909	0.903	0.900	0.906
16	1.000	0.941	0.940	0.940	0.938	0.931	0.927	0.931
17	1.000	0.962	0.956	0.957	0.955	0.947	0.942	0.949
18	1.000	0.939	0.932	0.928	0.925	0.917	0.913	0.916
19	1.000	0.939	0.933	0.929	0.928	0.923	0.919	0.923
20	1.000	0.951	0.948	0.941	0.943	0.937	0.932	0.933
Average	1.0000	0.9426	0.9364	0.9344	0.9322	0.9254	0.9208	0.9241
ln(Average)	0.00000	-0.05911	-0.06571	-0.06785	-0.07021	-0.07753	-0.08251	-0.07893

The results of the least-squares fit using this dataset are shown in **Table E-9** for case temperature of 55 °C and in **Table E-10** for 85 °C. It should be noted that in this

example, following the instructions in **Section 5.2.3**, the data from 6,000 hours to 10,000 hours are used for calculations of projected luminous flux maintenance life.

Table E-9. Least-Squares Curve-Fit for 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Test Data at Temperature Point $T_{s,1} = 55\text{ }^{\circ}\text{C}$ Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Point #	Time [h]	ln(Average)	xy	x	y	x ²
1	5,000	-0.04573	-228.7	5,000	-0.0457	2.500E+07
2	6,000	-0.04688	-281.3	6,000	-0.0469	3.600E+07
3	7,000	-0.04604	-322.3	7,000	-0.0460	4.900E+07
4	8,000	-0.05087	-407.0	8,000	-0.0509	6.400E+07
5	9,000	-0.05477	-492.9	9,000	-0.0548	8.100E+07
6	10,000	-0.05182	-518.2	10,000	-0.0518	1.000E+08
	Sums	-0.29611	-2250.3	45,000	-0.2961	3.550E+08
	Slope	-1.684E-06				
	Intercept	-3.672E-02				
	Reported α_1	2.000E-06				
	B_1	9.639E-01				
	Reported R_{85} [350, 850]	>60,000				

Table E-10. Least-Squares Curve-Fit for 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Test Data at Temperature Point $T_{s,1} = 85\text{ }^{\circ}\text{C}$ Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Point #	Time [h]	ln(Average)	xy	x	y	x ²
1	5,000	-0.06571	-328.6	5,000	-0.0657	2.500E+07
2	6,000	-0.06785	-407.1	6,000	-0.0679	3.600E+07
3	7,000	-0.07021	-491.5	7,000	-0.0702	4.900E+07
4	8,000	-0.07753	-620.2	8,000	-0.0775	6.400E+07
5	9,000	-0.08251	-742.6	9,000	-0.0825	8.100E+07
6	10,000	-0.07893	-789.3	10,000	-0.0789	1.000E+08
	Sums	-0.44274	-3379.3	45,000	-0.4427	3.550E+08
	Slope	-3.354E-06				
	Intercept	-4.863E-02				
	Reported α_1	3.354E-06				
	B_1	9.525E-01				
	Reported R_{85} [350, 850]	33,900				

E.4 Example of Arrhenius Interpolation Using 10,000 Hours of ANSI/IES LM-80-15 Radiant Flux Data

If the in-situ case temperature is $T_{s,i} = 70^\circ\text{C}$, the data in **Table E-9** ($T_{s,1} = 55^\circ\text{C}$) and **Table E-10** ($T_{s,2} = 85^\circ\text{C}$) are used to interpolate, and to project the radiant flux maintenance life for $T_{s,i} = 70^\circ\text{C}$. The parameters are shown as in **Table E-11**.

Table E-11. Parameters of Interpolation Using 10,000 Hours ANSI/IES LM-80-15 Radiant Flux Data for In-Situ Case Temperature $T_{s,i} = 70^\circ\text{C}$ Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Parameter	Temperature		
	Condition 1	Condition 2	Interpolation
$T_s (^\circ\text{C})$	55	85	70
$T_s (K)$	328.15	358.15	343.15
α	2.000E-06	3.354E-06	2.620E-06
B	0.9639	0.9525	0.9582
E_a/k_B	2026	N.A.	N.A.
A	9.602E-04	N.A.	N.A.
m_B	-3.800E-04	N.A.	N.A.
b_B	0.9848	N.A.	N.A.
Reported $R_{85} [350, 850]$	>60,000	33,900	45,700

The graphic representation of the radiant flux maintenance life projection results for $T_{s,1} = 55^\circ\text{C}$, $T_{s,2} = 85^\circ\text{C}$, and in-situ temperature of $T_{s,i} = 70^\circ\text{C}$ is shown in **Figure E-2**, for 10,000 hours of ANSI/IES LM-80-15 data.

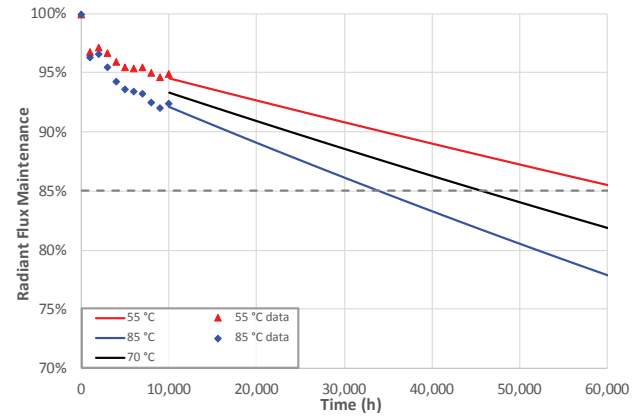


Figure E-2. Graphic representation of radiant flux maintenance life projection using 10,000 hours of ANSI/IES LM-80-15 data for temperature interpolation evaluated over the wavelength interval of 350 nm to 850 nm.

The numerical representation of the radiant flux maintenance life projection results for $T_{s,1} = 55^\circ\text{C}$, $T_{s,2} = 85^\circ\text{C}$, and in-situ temperature of $T_{s,i} = 70^\circ\text{C}$ is shown in **Table E-12**.

Table E-12. Numerical Results of Radiant Flux Maintenance Life Projection Using 10,000 Hours of ANSI/IES LM-80-15 Data Evaluated Over the Wavelength Interval of 350 nm to 850 nm

Time (h)	Projected Radiant Flux Maintenance		
	55 °C	70 °C	85 °C
10,000	0.9448	0.9334	0.9211
11,000	0.9429	0.9310	0.9180
12,000	0.9410	0.9285	0.9149
13,000	0.9392	0.9261	0.9119
14,000	0.9373	0.9237	0.9088
15,000	0.9354	0.9213	0.9058
16,000	0.9335	0.9189	0.9027
17,000	0.9317	0.9165	0.8997
18,000	0.9298	0.9141	0.8967
19,000	0.9280	0.9117	0.8937
20,000	0.9261	0.9093	0.8907
21,000	0.9243	0.9069	0.8877
22,000	0.9224	0.9045	0.8847
23,000	0.9206	0.9022	0.8818
24,000	0.9187	0.8998	0.8788
25,000	0.9169	0.8975	0.8759
26,000	0.9151	0.8951	0.8730
27,000	0.9132	0.8928	0.8700
28,000	0.9114	0.8904	0.8671
29,000	0.9096	0.8881	0.8642
30,000	0.9078	0.8858	0.8613
31,000	0.9060	0.8835	0.8584
32,000	0.9041	0.8812	0.8556
33,000	0.9023	0.8788	0.8527
34,000	0.9005	0.8765	0.8498
35,000	0.8987	0.8743	0.8470

Time (h)	Projected Radiant Flux Maintenance		
	55 °C	70 °C	85 °C
36,000	0.8969	0.8720	0.8442
37,000	0.8951	0.8697	0.8413
38,000	0.8934	0.8674	0.8385
39,000	0.8916	0.8651	0.8357
40,000	0.8898	0.8629	0.8329
41,000	0.8880	0.8606	0.8301
42,000	0.8862	0.8584	0.8273
43,000	0.8845	0.8561	0.8246
44,000	0.8827	0.8539	0.8218
45,000	0.8809	0.8516	0.8191
46,000	0.8792	0.8494	0.8163
47,000	0.8774	0.8472	0.8136
48,000	0.8757	0.8450	0.8109
49,000	0.8739	0.8428	0.8081
50,000	0.8722	0.8406	0.8054
51,000	0.8704	0.8384	0.8027
52,000	0.8687	0.8362	0.8000
53,000	0.8670	0.8340	0.7974
54,000	0.8652	0.8318	0.7947
55,000	0.8635	0.8296	0.7920
56,000	0.8618	0.8275	0.7894
57,000	0.8600	0.8253	0.7867
58,000	0.8583	0.8231	0.7841
59,000	0.8566	0.8210	0.7815
60,000	0.8549	0.8188	0.7789

E.5 Example of Current Interpolation Using 9,000 Hours of ANSI/IES LM-80-15 Luminous Flux Data

Table E-13 represents the data sets from 20 units of tested samples for 9,000 hours of ANSI/IES LM-80-15 test at drive currents of 280 mA and 320 mA. Individual device data have been omitted for clarity.

Table E-13. ANSI/IES LM-80-15 Test Data at $I_{f,1} = 280$ mA and $I_{f,2} = 320$ mA

Time (h)	Average Luminous Flux	
	280 mA	320 mA
0	1.0000	1.0000
1,000	0.9663	0.9483
2,000	0.9384	0.9189
3,000	0.9122	0.8868
4,000	0.8986	0.8544
5,000	0.8800	0.8295
6,000	0.8589	0.7799
7,000	0.8283	0.7331
8,000	0.8081	0.7011
9,000	0.7971	0.6088

Using the least-squares fit procedure outlined in **Annex C**, the slope, intercept, α , and B values are calculated for both conditions, as shown in **Table E-14**.

Table E-14. Least-Squares Fit Data and Parameters Calculated from the Data in Table E-13.

Parameter	280 mA	320 mA
Slope	-2.546E-05	-6.460E-05
Intercept	-3.657E-03	1.274E-01
Reported α	2.546E-05	6.460E-05
B	0.9963	1.1359

If the drive current is $I_{f,i} = 285$ mA, the data in **Table E-14** ($I_{f,1} = 280$ mA and $I_{f,2} = 320$ mA) are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 285$ mA. The parameters are shown as in **Table E-15**.

The graphic representation of the luminous flux maintenance life projection results for $I_{f,1} = 280$ mA, $I_{f,2} = 320$ mA, and a drive current of $I_{f,i} = 285$ mA are shown in **Figure E-3** for 9,000 hours of ANSI/IES LM-80-15 data.

Table E-15. Parameters of Interpolation Using 9,000 hours of ANSI/IES LM-80-15 Data for Drive Current $I_{f,i} = 285$ mA

Parameter	Current (mA)		
	280	320	285
α	2.546E-05	6.460E-05	3.037E-05
B	0.9963	1.1359	1.0138
m_α	9.785E-07	N.A.	N.A.
b_α	-2.485E-04	N.A.	N.A.
m_B	3.490E-03	N.A.	N.A.
b_B	0.01910	N.A.	N.A.
Reported L_{70}	13,900	8,000*	12,200

Table note:

* During testing, L_{70} occurred between 8,000 hours and 9,000 hours. The interpolated value of L_{70} is 8,012 hours, which is rounded to 8,000 hours.

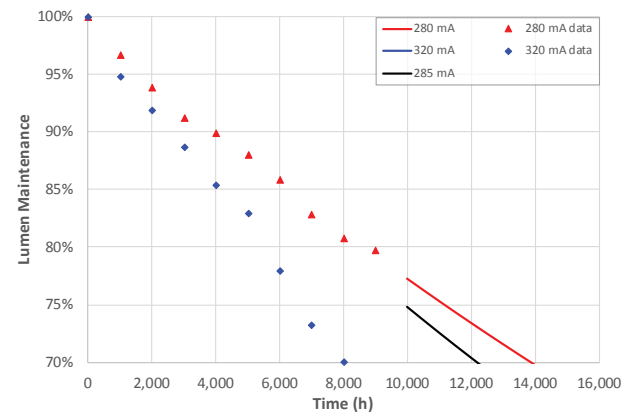


Figure E-3. Graphic representation of luminous flux maintenance life projection using 9,000 hours of ANSI/IES LM-80-15 data for current interpolation.

The numerical representation of the luminous flux maintenance life projection results for $I_{f,1} = 280$ mA, $I_{f,2} = 320$ mA, and a drive current of $I_{f,i} = 285$ mA are shown in **Table E-16**.

Table E-16. Numerical Results of Luminous Flux Maintenance Life Projection Using 9,000 Hours of ANSI/IES LM-80-15 Data

Time (h)	Projected Luminous Flux Maintenance		
	280 mA	320 mA	285 mA
10,000	0.7724	0.5954	0.7482
11,000	0.7529	0.5581	0.7258
12,000	0.7340	0.5232	0.7041
13,000	0.7156	0.4905	0.6831
14,000	0.6976	0.4598	0.6626

E.6 Example of Current Interpolation Using 7,000 Hours of ANSI/IES LM-80-15 Data

Table E-17 represents the data sets from 20 units of tested samples for 7,000 hours of ANSI/IES LM-80-15 testing at drive currents of 80 mA and 120 mA. Individual device data have been omitted for clarity.

Table E-17. ANSI/IES LM-80-15 Luminous Flux Data at $I_{f,1} = 80$ mA and $I_{f,2} = 120$ mA

Time (h)	Average Luminous Flux	
	80 mA	120 mA
0	1.0000	1.0000
1,000	0.9818	0.9938
2,000	0.9773	0.9718
3,000	0.9701	0.9633
4,000	0.9615	0.9311
5,000	0.9529	0.9134
6,000	0.9491	0.8909
7,000	0.9392	0.8763

Using the least-squares fit procedure outlined in **Annex C**, the slope, intercept, α , and B values are calculated for both conditions, as shown in **Table E-18**.

Table E-18. Least-Squares Fit Data and Parameters Calculated from the Data in Table E-17

Parameter	80 mA	120 mA
Slope	-7.813E-06	-2.202E-05
Intercept	-7.472E-03	1.985E-02
Reported α	7.813E-06	2.202E-05
B	0.9926	1.0200

If the drive current is $I_{f,i} = 95$ mA, the data in **Table E-18** ($I_{f,1} = 80$ mA and $I_{f,2} = 120$ mA) are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 95$ mA. The parameters are shown as in **Table E-19**.

The graphic representation of the luminous flux maintenance life projection results for $I_{f,1} = 80$ mA, $I_{f,2} = 120$ mA, and a drive current of $I_{f,i} = 95$ mA are shown in **Figure E-4**, for 7,000 hours of ANSI/IES LM-80-15 data.

Table E-19. Parameters of Interpolation Using 7,000 Hours of ANSI/IES LM-80-15 Luminous Flux Data for Drive Current $I_{f,i} = 95$ mA

Parameter	Current (mA)		
	80	120	95
α	7.813E-06	2.202E-05	1.314E-05
B	0.9926	1.0200	1.0029
m_α	3.5518E-07	N.A.	N.A.
b_α	-2.060E-05	N.A.	N.A.
m_B	6.873E-04	N.A.	N.A.
b_B	0.9376	N.A.	N.A.
Reported L_{70}	>42,000	17,100	27,400

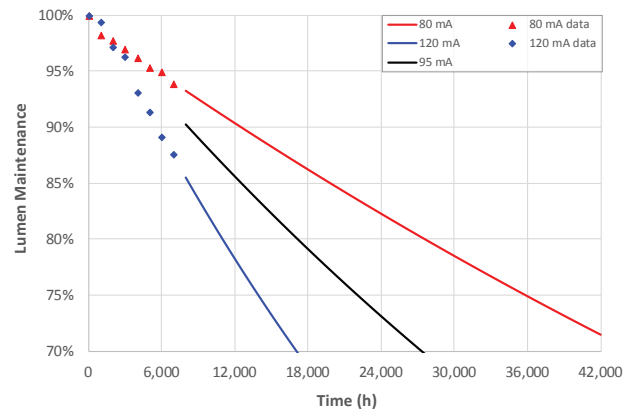


Figure E-4. Graphic representation of luminous flux maintenance life projection using 7,000 hours of ANSI/IES LM-80-15 luminous flux data for current interpolation.

E.7 Example of Arrhenius Interpolation and Current Interpolation Using 6,000 Hours of ANSI/IES LM-80-15 Luminous Flux Data

If the drive current is 90 mA and the in-situ case temperature is $T_{s,i} = 90$ °C, the data in **Table E-20** ($I_{f,1,1} = 80$ mA, $I_{f,1,2} = 150$ mA, $T_{s,1} = 85$ °C, $I_{f,2,1} = 80$ mA, $I_{f,2,2} = 150$ mA, $T_{s,2} = 105$ °C) are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 95$ mA and $T_{s,i} = 90$ °C. Individual device data have been omitted for clarity. The parameters are shown as in **Table E-21**.

Using the least-squares fit procedure outlined in **Annex C**, the slope, intercept, α , and B values are calculated for all four conditions, as shown in **Table E-19**.

Table E-20. ANSI/IES LM-80-15 Luminous Flux Data at $I_{f1,1} = 80$ mA, $I_{f1,2} = 150$ mA, and $T_{s,1} = 85$ °C; and $I_{f2,1} = 80$ mA, $I_{f2,2} = 150$ mA, and $T_{s,2} = 105$ °C

Time, h	Average Luminous Flux			
	85 °C (358.15 K)		105 °C (378.15)	
	80 mA	150 mA	80 mA	150 mA
0	1.0000	1.0000	1.0000	1.0000
1,000	0.9863	0.9862	0.9839	0.9741
2,000	0.9808	0.9830	0.9708	0.9641
3,000	0.9761	0.9800	0.9616	0.9578
4,000	0.9736	0.9787	0.9549	0.9544
5,000	0.9718	0.9774	0.9491	0.9511
6,000	0.9700	0.9760	0.9436	0.9485

These parameters are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 95$ mA and $T_{s,i} = 90$ °C. The parameters are shown as in **Table E-22**.

Table E-21. Least-Squares Fit Data and Parameters Calculated from the Data in Table E-20

Parameter	85 °C		105 °C	
	80 mA	150 mA	80 mA	150 mA
Slope	-3.244E-06	-2.013E-06	-8.112E-06	-5.070E-06
Intercept	-1.251E-02	-1.294E-02	-1.185E-02	-2.486E-02
Reported α	3.244E-06	2.013E-06	8.112E-06	5.070E-06
B	0.9876	0.9871	0.9882	0.9754
Reported L_{70}	>36,000	>36,000	>36,000	>36,000

The graphic representation of the luminous flux maintenance life projection results for $I_{f1,1} = 80$ mA, $I_{f1,2} = 150$ mA, $T_{s,1} = 85$ °C; and $I_{f2,1} = 80$ mA, $I_{f2,2} = 150$ mA, $T_{s,2} = 105$ °C; and a drive current of $I_{f,i} = 95$ mA and $T_{s,i} = 90$ °C are shown in **Figure E-5**, for 6,000 hours of ANSI/IES LM-80-15 luminous flux data.

Table E-22. Parameters of Interpolation Using 6,000 Hours of ANSI/IES LM-80-15 Luminous Flux Data for Drive Current $I_{f,i} = 90$ mA and In-Situ Case Temperature $T_{s,i} = 90$ °C

		Parameter	Conditions				
			1, 1	1, 2	2, 1	2, 2	Interpolation
Data		Temperature (°C)	85	85	105	105	90
		Temperature (K)	358.15	358.15	378.15	378.15	363.15
		Current (mA)	80	150	80	150	95
		α	3.244E-06	2.013E-06	8.112E-06	5.070E-06	3.784E-06
		B	0.9876	0.9871	0.9882	0.9754	0.9877
Interpolation	Current	m_α	-1.759E-08	N.A.	-4.346E-08	N.A.	N.A.
		b_α	4.651E-06	N.A.	1.159E-05	N.A.	N.A.
		m_B	-6.065E-06	N.A.	-1.825E-04	N.A.	N.A.
		b_B	0.9881	N.A.	1.0028	N.A.	N.A.
		α	2.980E-06	N.A.	7.460E-06	N.A.	N.A.
		B	0.9876	N.A.	0.9882	N.A.	N.A.
		Reported L_{70}	>36,000	N.A.	>36,000	N.A.	>36,000
	Temperature	E_d/k_B	6,214	N.A.	N.A.	N.A.	N.A.
		A	1.021E+02	N.A.	N.A.	N.A.	N.A.
		m_B	3.260E-05	N.A.	N.A.	N.A.	N.A.
		b_B	0.9848	N.A.	N.A.	N.A.	N.A.
		α	3.784E-06	N.A.	N.A.	N.A.	N.A.
		B	0.9877	N.A.	N.A.	N.A.	N.A.
		Reported L_{70}	>36,000	N.A.	N.A.	N.A.	N.A.

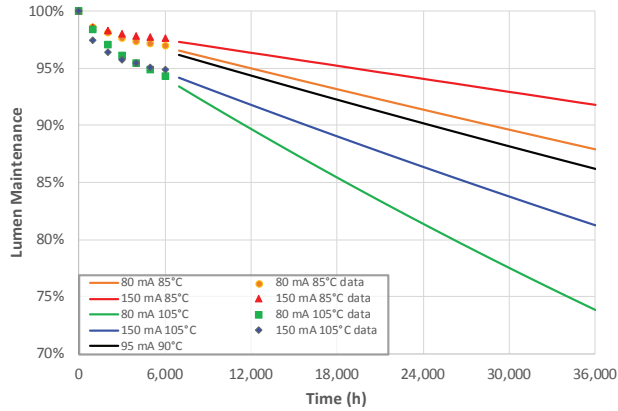


Figure E-5. Graphic representation of luminous flux maintenance life projection using 6,000 hours of ANSI/IES LM-80-15 luminous flux data for simultaneous current and temperature interpolation.

E.8 Example of Arrhenius Interpolation and Current Interpolation Using 6,000, 7,000, and 8,000 hours of ANSI/IES LM-80-15 Luminous Flux Data

If the drive current is 920 mA and the in-situ case temperature is $T_{s,i} = 98^\circ\text{C}$, the data in **Table E-23** ($I_{f1,1} = 700$ mA, $I_{f1,2} = 1,200$ mA, $T_{s,1} = 85^\circ\text{C}$, $I_{f2,1} = 700$ mA, $I_{f2,2} = 1,000$ mA, $T_{s,2} = 105^\circ\text{C}$) are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 920$ mA and $T_{s,i} =$

Table E-23. ANSI/IES LM-80-15 Luminous Flux Data at $I_{f1,1} = 80$ mA, $I_{f1,2} = 150$ mA, and $T_{s,1} = 85^\circ\text{C}$; and $I_{f2,1} = 80$ mA, $I_{f2,2} = 150$ mA, and $T_{s,2} = 105^\circ\text{C}$

Time (h)	Average Luminous Flux			
	85 °C (358.15 K)		105 °C (378.15 K)	
	700 mA	1,200 mA	700 mA	1,000 mA
0	1.0000	1.0000	1.0000	1.0000
500	0.9936	0.9646	0.9718	0.9492
1,000	0.9852	0.9496	0.9572	0.9326
2,000	0.9798	0.9420	0.9492	0.9230
3,000	0.9686	0.9306	0.9356	0.9116
4,000	0.9628	0.9246	0.9286	0.9054
5,000	0.9608	0.9220	0.9268	0.8942
6,000	0.9562	0.9058	0.9210	0.8846
7,000	0.9512	0.8908	0.9070	
8,000	0.9430			

98°C . Individual device data have been omitted for clarity. The parameters are shown as in **Table E-24**.

Using the least-squares fit procedure outlined in **Annex C**, the slope, intercept, α , and B values are calculated for all four conditions, as shown in **Table E-24**.

Table E-24. Least-Squares Fit Data and Parameters Calculated from the Data in Table E-23

Parameter	85 °C	85 °C	105 °C	105 °C
	700 mA	1,200 mA	700 mA	1,000 mA
Slope	-5.003E-06	-1.038E-05	-7.900E-06	-1.046E-05
Intercept	-1.637E-02	-3.760E-02	-3.923E-02	-5.943E-02
Reported α	5.003E-06	1.038E-05	7.900E-06	1.046E-05
B	0.9838	0.9631	0.9615	0.9423
Reported L_{70}	>48,000	30,700	40,200	28,400

Because the dataset for 700 mA and 85 °C extended to 8,000 hours, extrapolation limits for that condition are 48,000 hours. Since the datasets for 1,200 mA and 85 °C, as well as for 700 mA and 105°C, extended to 7,000 hours, the maximum extrapolation limits for those conditions would be 42,000 hours. The final dataset for 1,000 mA and 105°C only extended to 6,000 hours, and it would have a maximum extrapolation limit of 36,000

hours. The minimum extrapolation limit for a dataset used in an interpolation applies to the outcome of that interpolation, as seen below.

These parameters are used to interpolate, and to project the luminous flux maintenance life for $I_{f,i} = 920$ mA and in-situ case temperature $T_{s,i} = 98$ °C. The parameters are shown as in **Table E-25**.

Table E-25. Parameters of Interpolation Using 6,000, 7,000 and 8,000 Hours of ANSI/IES LM-80-15 Luminous Flux Data for Drive Current $I_{f,i} = 920$ mA and In-Situ Case Temperature $T_{s,i} = 98$ °C

			Conditions				
			1, 1	1, 2	2, 1	2, 2	Interpolation
	Data	Temperature (°C)	85	85	105	105	98
		Temperature (K)	358.15	358.15	378.15	378.15	371.15
		Current (mA)	700	1,200	700	1,000	920
		α	5.003E-06	1.038E-05	7.900E-06	1.046E-05	8.886E-06
		B	0.9838	0.9631	0.9615	0.9423	0.9693
Interpolation	Current	m_α	1.075E-08	N.A.	8.530E-09	N.A.	N.A.
		b_α	-2.522E-06	N.A.	1.929E-06	N.A.	N.A.
		m_B	-4.133E-05	N.A.	-6.409E-05	N.A.	N.A.
		b_B	1.0127	N.A.	1.0064	N.A.	N.A.
		α	7.368E-06	N.A.	9.777E-06	N.A.	N.A.
		B	0.9838	N.A.	0.9615	N.A.	N.A.
		Reported L_{70}	>42,000	N.A.	32,500	N.A.	>36,000
	Temperature	E_α/k_B	1915	N.A.	N.A.	N.A.	N.A.
		A	1.549E-03	N.A.	N.A.	N.A.	N.A.
		m_B	-1.112E-03	N.A.	N.A.	N.A.	N.A.
		b_B	1.0783	N.A.	N.A.	N.A.	N.A.
		α	8.886E-06	N.A.	N.A.	N.A.	N.A.
		B	0.9693	N.A.	N.A.	N.A.	N.A.
Reported L_{70}		>36,000	N.A.	N.A.	N.A.	N.A.	

The first current interpolation used the dataset for 700 mA and 85 °C (total duration 8,000 hours) and the dataset for 1,200 mA and 85 °C (total duration 7,000 hours). Interpolated conditions inherit the minimum extrapolation limit from their bounding conditions, in this case the 42,000-hour limit imposed by the dataset for 1,200 mA and 85 °C.

The final interpolation included the dataset for 1,000 mA and 105 °C (total duration 6,000 hours) and was therefore has a maximum extrapolation limit of 36,000 hours.

The graphic representation of the luminous flux maintenance life projection results for $I_{f1,1} = 700$ mA, $I_{f1,2} = 1,200$ mA, $T_{s,1} = 85$ °C; and $I_{f2,1} = 700$ mA, $I_{f2,2} = 1,000$ mA, $T_{s,2} = 105$ °C; and a drive current of $I_{f,i} = 920$ mA and in-situ case temperature $T_{s,i} = 98$ °C are shown in **Figure E-6**.

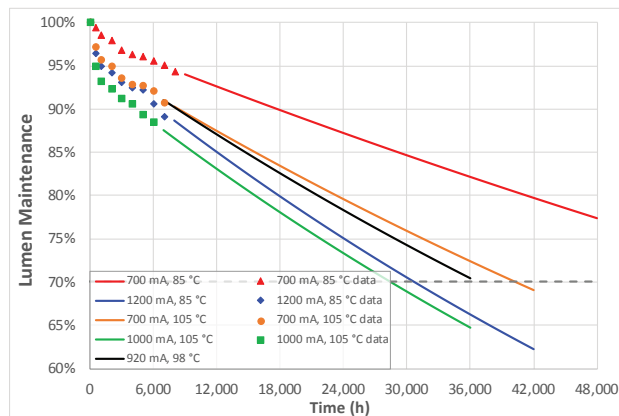


Figure E-6. Graphic representation of luminous flux maintenance life projection using 6,000, 7,000 and 8,000 hours of ANSI/IES LM-80-15 data for simultaneous current and temperature interpolation.

Annex F – Consideration of Manufacturer’s Prediction Model

In developing and producing LED packages, manufacturers have used different technologies for substrates, structures, encapsulation and lens materials, phosphors, and other components. The LEDs produced by different manufacturers demonstrated vast differences in the lumen degradation behaviors. The degradations were accelerated with the severity of the

operation and environment conditions. In general, the acceleration factors can be: a) temperature induced acceleration; b) current density induced acceleration; c) optical radiation induced acceleration; d) humidity induced acceleration; e) combination of optical radiation and temperature induced acceleration; f) others. Based on the analytical approaches and testing data collections, LED manufacturers have developed mathematical models to predict their products’ luminous flux maintenance lives. The model can be a mathematical equation or a polynomial, and parameters used in the mathematical expression may be chosen based on experimental or engineering judgments.

The original IES TM-21 Working Group (WG) has considered adopting the prediction models provided by LED manufacturers. The proposed approach was to request that the manufacturer provide the mathematical model, the test data, and the luminous flux maintenance life prediction. Then the document users should be able to verify the validity of the manufacturer’s model by comparing to the simple exponential model using the same data provided by the manufacturer, and identifying that the manufacturer’s model has higher modeling and projection accuracy. Therefore, the WG needed to develop a consistent and reliable method to make the above verification and to approve the validity of the LED manufacturers’ predictions. That was important and necessary to ensure the integrity of the document.

During almost three years in developing the original version of this document, the WG conducted statistical analyses for over 40 sets of IES LM-80-08 (the predecessor of IES LM-80-15) test data collected from four major LED manufacturers, and more than 20 sets data were collected with testing duration over 10,000 hours. To determine whether a manufacturer model fits the data statistically better than the model described in **Section 5** (simple exponential model), the RMSE (root-mean-square error) and other measures were used in the analyses. However, the WG did not find a reliable measure to distinguish which model—manufacturer or simple exponential model—appears to provide more accurate prediction using 6,000 or up to 10,000 hours’ data. Therefore, the WG decided that further studies are needed for finding a method to validate a model provided by LED manufacturers.

Annex G – Analysis of Mathematical Modeling as a Method of Projecting Luminous Flux Maintenance Life

References for Annex G are provided in **Section G.5**.

G.1 Analysis Approach

An initial approach to the problem of projecting luminous flux maintenance life was the consideration of multiple mathematical models. It is known that the various parts of a typical LED product can affect the lumen output over time, and if these effects could be mathematically characterized, this might provide a useful method of projection. An initial set of models was suggested that relate to the physics of lumen decay for various parts and/or operating conditions. Additional models and combinations of models were added during the analysis.

All of the early research work in modeling to explain LED degradation was based on semiconductor physics (reactions and motion of various defects). One researcher¹ attributed the degradation to the drift of charged point defects in the quasi-neutral regions of a crystal, leading to a linear dependence with time under certain conditions. Other analysis² introduced the simple exponential model for LED light decay with time. Another model³ was based on quasi-chemical reactions of defects, leading to exponential, faster-than-exponential, or slower-than-exponential theoretical decay curves. The work by Ptashchenko⁴ defined the degradation rate of LEDs by:

$$S = \left(\frac{d\Phi}{dt} \right) / \Phi, \quad (\text{G-1})$$

where Φ is LED flux output.

The more recent advent of high brightness white LEDs suitable for general illumination generated issues regarding LED luminous flux maintenance over time, with related published studies.⁵⁻⁹ Some of the studies^{5,8} highlight the importance of degradation of subcomponents other than the chip in an LED package. The encapsulant is especially considered important. Thermal degradation of the encapsulant leading to lumen loss has been observed in high-temperature

storage experiments.⁹ An empirical exponential fit of the lumen decay curve over operating time (excluding the first 1,000 hours) has been proposed.⁵ Results show a large variation in lumen decay rate among different packages, attributable to the use of different heat extraction techniques and materials. The final set of models considered in the analysis is listed in **Table G-1**, which summarizes the decay parameters, mathematical solutions, and degradation mechanism bases for the decay models.

In the decay rate models, the k_1 term is based on the expectation of linear luminous flux dependence over time for some cases. The k_2 term can be obtained by setting $S = k_2$ in **Equation G1**. The k_3 term is introduced in order to account for possible oxidation or corrosion effects of metals used to make reflectors around or under the chip in LED packages. These effects are believed to follow a logarithmic law in some cases.¹⁰ Terms are also combined in the rate equations, to explore mixed decay scenarios (Models 3, 5, and 8).

The approach to considering these models involved looking at the potential fit of the various models to various different types of known and expected LED lumen decay data. By categorizing and comparing the goodness of fit, it was thought that a model or set of models could be identified as most effectively representing expected lumen output decay, given the data, and therefore be used to project luminous flux maintenance life.

Standard statistical criteria such as coefficient of determination (R^2), the sum of squared errors of prediction (SSE)* and root-mean-squared error ($RMSE$) are generally used to gauge the “goodness-of-fit of models.” These criteria measure the differences between values predicted by a model and the values actually observed. The individual differences, called *residuals*, are aggregated together into a single measure of predictive power. The benefit of $RMSE$ over SSE is that $RMSE$ accounts for the relationship of the number of parameters (p) in the model to the number of data observations (n), as shown in **Equation G-2**. $RMSE$ more accurately accounts for model complexity.

* Sometimes called *residual sum of squares* (RSS).

Table G-1. Engineering-Based Models Used in the Analysis of Model Fitting for LED Lumen Decay Life Projection.

Model	Decay Rate	Closed Form Solution	Comment
1	$\frac{dI_v}{dt} = k_1$	$I_v = I_v^0 + k_1(t - t^0)$	
2	$\frac{dI_v}{dt} = k_2 I_v$	$I_v = I_v^0 \exp[k_2(t - t^0)]$	
3	$\frac{dI_v}{dt} = k_1 I_v + k_2 I_v$	$I_v = \left(I_v^0 + \frac{k_1}{k_2}\right) \exp[k_2(t - t^0)] - \frac{k_1}{k_2}$	Model 1 + Model 2
4	$\frac{dI_v}{dt} = \frac{k_3}{t}$	$I_v = I_v^0 + k_3 \ln\left(\frac{t}{t^0}\right)$	
5	$\frac{dI_v}{dt} = k_1 + \frac{k_3}{t}$	$I_v = I_v^0 + k_1(t - t^0) + k_3 \ln\left(\frac{t}{t^0}\right)$	Model 1 + Model 4
6	$\frac{dI_v}{dt} = -k_4 I_v^2$	$I_v = \frac{I_v^0}{1 + I_v^0 k_4(t - t^0)}$	
7	$\frac{dI_v}{dt} = k_5 \frac{I_v}{t}$	$I_v = I_v^0 \left(\frac{t}{t^0}\right)^{k_5}$	
8	$\frac{dI_v}{dt} = k_2 I_v + \frac{k_5 I_v}{t}$	$I_v = I_v^0 \exp[k_2(t - t^0)] \left(\frac{t}{t^0}\right)^{k_5}$	Model 2 + Model 7
9		$I_v = I_v^0 \exp\left[-\frac{(t - t^0)^{k_6}}{k_6}\right]$	

$$RMSE(\theta) = \sqrt{MSE(\theta)} = \sqrt{\frac{SSE(\theta)}{n - p}} \quad (\text{G-2})$$

The preliminary study results indicate that for most LED decay data, the differences of R^2 , SSE , and $RMSE$ among many models are not significant enough to suggest an absolute best fitting model. Furthermore, uncertainty (noise) is inherent in measurement data due to, for example, apparatus accuracy, time dependent factors, and measurement repeatability. Initial statistical analyses on the effects of data noise suggest that even a small amount of data noise could result in poor model differentiation and large life projection variability. Therefore, to gain insight in the behaviors of the various mathematical models for LED degradation, life projection, and effects of various test

related conditions, a statistical analysis approach was developed.

G.2 Theoretical and Real Data

To cleanly but also realistically identify the fit of various models, statistical analyses were conducted on both theoretical data and real LED degradation data obtained from manufacturers.

Theoretical data in the form of seven datasets that emulate typical LED decay behaviors were constructed to represent the potential variety of LED decay data and are presented in **Figure G-1**. The “candy cane” scenario exhibits a hump in the initial data that characterizes a transient effect in the LED lumen output during the “warming up” period. Linear and curvilinear scenarios represent two other LED decay profiles commonly

manifested in available LED test data. Accelerated decay is observed with some LED products, in which the lumen output drops quickly. In the limited minimum test period defined in IES LM-80-15, lumen output of some LED products were not permitted to show any sign of lumen depreciation, some lumen output may flatten out after decreasing for a while, and some may rise again above the earlier lumen output after initial decrease. These are captured in the flat, asymptotic, and U-shaped scenarios respectively. To make these simulation results applicable to real data taken using the ANSI/IES LM-80-15 test method, theoretical data were designed with a 1,000-hour interval and 6,000-hour total duration. In most of these analyses, 500 simulated data sets were generated by adding 1% noise to represent laboratory equipment and measurement uncertainties. This 1% noise level is considered appropriate by taking into account both measurement system and ambient uncertainty budgets over time and is corroborated by work done on measurement uncertainty at the standard testing lab at the U. S. National Institute of Standards and Technology (NIST). The constitution of the data is described in **Section G.3** and its subsections.

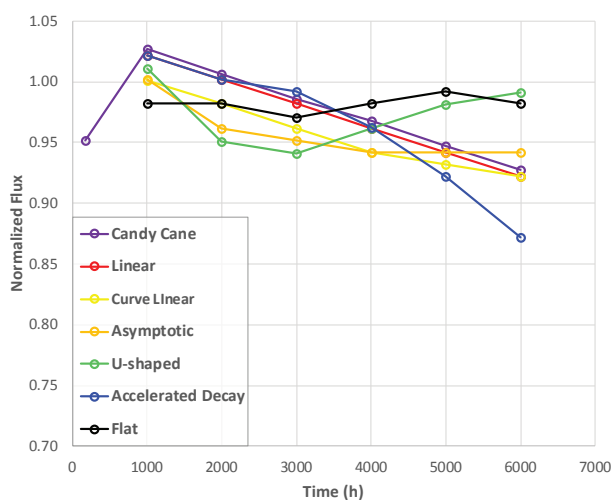


Figure G-1. Theoretical decay scenarios reflecting typical and/or possible LED light source decay profiles.

Appropriate exploration of the model selection behavior also included using decay data from real LED products obtained from LED manufacturers with product and manufacturers' names and detailed operation conditions removed for unbiased analysis.

These decay data were measured using the ANSI/IES LM-80-15 test method, with some data sets containing additional measurements taken at shorter intervals (less than every 1,000 hours) and/or beyond 6,000 hours. All data were unprocessed prior to analysis, with the exception of being normalized at the first data point. Decay data of 29 different LED products from various manufacturers were obtained for this study. Among them, the sample size of each product ranged from 6 to 30 samples.

G.3 Analysis and Results

The models listed in **Table G-1** were programmed in the statistical software "R" and fit to each simulated dataset described above. Evaluations were made on the set of simulations based on analyzing the statistics of the goodness-of-fit for each model. L_{70} projection distributions or prediction bands were generated by combining the model standard errors of all simulations for each model. The subsections below describe the detailed analysis and results for each data type.

G.3.1 Results from Theoretical Data. In the study of theoretical decay data (see **Figure G-1**), the statistical distribution of RMSE values was chosen to present an overall picture of the goodness-of-fit for each model at a given set of decay data. Using the "candy cane" data as an example, as shown in **Figure G-2**, the x-axis is the model number (refer to **Table G-1** for model details), and each dot is a calculated RMSE value from each simulation. The "box and whisker" plot visually aids in describing the distribution of RMSE values, where the bottom and top lines of the box show the 25th and 75th percentiles of the RMSE values, respectively.

The decay profiles of the different models for the same dataset are plotted in **Figure G-3**. The L_{70} projection distribution for each model is expressed using a trimmed mean (thick line) and an error bar representing the 25th to 75th percentiles of the flux maintenance values on the y-axis of in **Figure G-3**. In this example, Model 5 (combination of linear and logarithm models) has significantly smaller RMSE values than the other models (see **Figure G-2**). It is the obvious selection as the "best fit" model for these data. Predicted flux values could be anywhere within the distribution calculated for Model

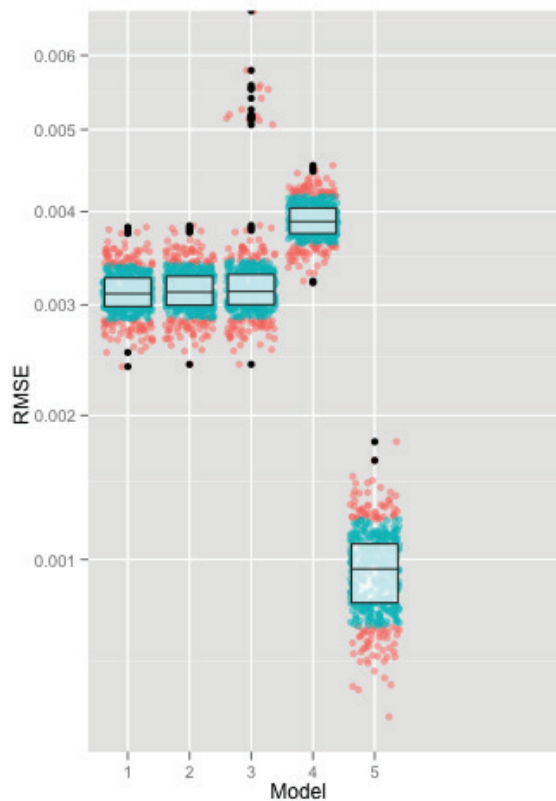


Figure G-2. Box plot of RMSE values for each model – “candy cane” data (see Figure G-1).

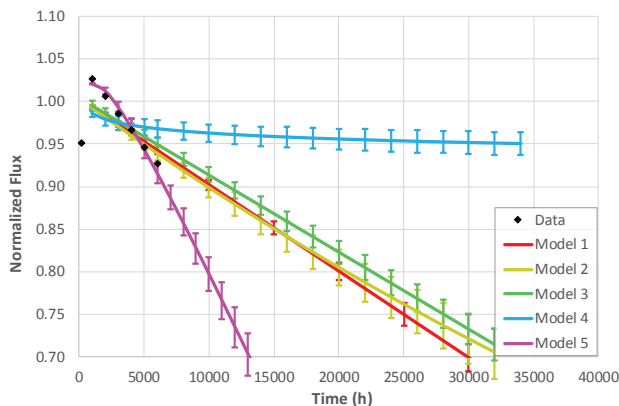


Figure G-3. Model projection – “candy cane” data, with time plotted in hours (see Figure G-1)

5. Therefore, a conservative approach to life projection would be the use of the lower, 25th percentile of the flux maintenance values as the final life projection for these data. That way, the uncertainty due to the fit is at least considered in an attempt to provide an estimate when using such drastic extrapolation.

G.3.2 Results from Real Data from LED Manufacturers. In many decay scenarios, the differences in RSME values among several models may not be sufficiently significant to pick an obvious winning model. In an example of a set of real LED decay data, shown in **Figure G-4**, the differences of RSME values among all models are not large enough to make a selection of the “best fit” model. A method applied in this study was to use the 10th and 90th percentiles of the two adjacent models as the criteria to separate models for model selection. A model is selected only if its 90th percentile is equal to or lower than the 10th percentile of the adjacent models.



Figure G-4. Box plot of RMSE values for each model – real LED data.

In this case a winning model could not be chosen under the aforementioned criteria as all models had similar RMSE results. The flux maintenance projection values calculated by the five models for this dataset are vastly different, as shown in **Figure G-5**. To be realistic in

LED life projection and protect consumers' interests, when none of the models stands out as the best fit, it is considered appropriate to choose the most conservative flux maintenance life prediction from those models with the lowest RMSE (those demonstrating appropriate model fit). In this case, Model 1 predicts the shortest flux maintenance life with the lower, 25th percentile of its distribution at approximately 36,000 hours. This LED product may have a potential maintenance lifetime longer than the projection of Model 1, but at the 6,000-hour test duration and resulting model fitting, there is insufficient evidence to provide an accurate longer prediction.

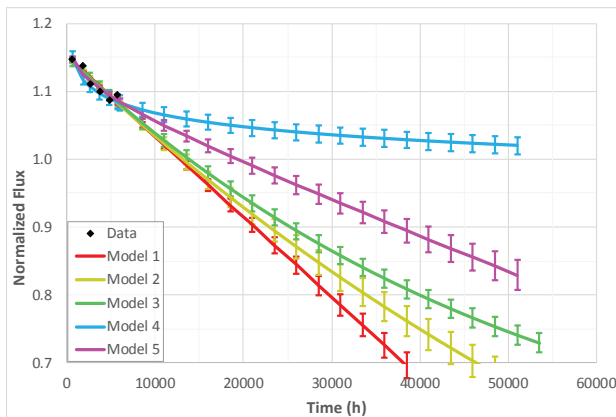


Figure G-5. Model projection – real LED data, with time plotted in hours.

The impact of noise from test condition variability on this small number of data points is detrimental to model fitting and always results in larger errors, which translates to greater variability in τ_{70} projections for a given model. Furthermore, the true decay profile of this particular LED may not yet have manifested itself after only 6,000 hours of data. When this same product was measured every 1,000 hours up to a total of 15,000 hours, the true decay profile was better represented by some of the models considered (see **Figure G-7**). This is shown in **Figure G-6**, where Models 3, 4, and 5 clearly fit the data better, as indicated by their lower RMSE values than those of Models 1 and 2. This would suggest that excluding Models 1 and 2, due to their lack of fit, from model selection would be appropriate. Because the overlaps of RMSE values between Models 3, 4 and 5 are greater than 10% of their populations respectively, one model among these does not dominate for τ_{70}



Figure G-6. Box plot of RMSE values – real LED data with longer test duration.

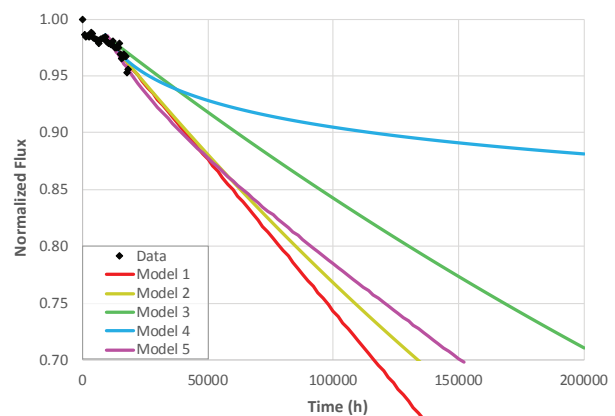


Figure G-7. Model projection – real LED data with longer test duration, with time plotted in hours.

estimates. Instead, the τ_{70} life prediction is chosen as the most conservative among this group. Model 5 has the smallest τ_{70} estimate of the three better fitting models, as seen in **Figure G-7**, and so the τ_{70} life of this LED product is considered to be 130,000 hours. In comparison, the results with only 6,000 hours of data

showed a much more variable distribution of the τ_{70} estimate, and a lower estimate of 36,000 hours. The available longer test data demonstrated an expected reduction in decay over time, and reduced variability in the model parameters, thus improving the argument for use of a goodness-of-fit criterion.

Additional modeling using the real product data was conducted on the longer datasets to attempt to validate an RMSE-based model choice, or elimination using a future actual data point. For this analysis, longer datasets (10,000 hours and greater) were fit to the first 6,000 hours (or longer if the datasets were longer than 10,000 hours). The predicted light output at the end of the data stream using the first 6,000 hours of data was compared with the actual output to determine whether the model(s) with the best RMSE fit consistently had the closest match to the real data point. Unfortunately, with the data sets available, this verification was not confirmed. The best model fit to the real data values was not always the model with the best RMSE fit at 6,000 hours, or in some cases up to 10,000 hours.

Other statistical techniques could be used to determine a “winning” model, including using the PRESS statistic.* However, such methods find only partial relevance to our problem. They are useful in predicting in the data space (region where data are available, such as 0 to 6,000 hours), but not generally applicable when extrapolating well beyond the limits of the data range.

G.4 Summary and Conclusions

The summary findings from these analyses pointed to the inability of a statistical metric such as RMSE (or SSE, or R^2) to always reasonably identify a model that could provide a “best fit” and therefore best projection (extrapolation) of LED lumen degradation. Specific conclusions include:

- RMSE (or R^2 , or SSE) comparing model fits will not be significant enough to pick a best model with only 6,000 hours of real data.

- RMSE may help identify a group of models at longer data periods, such as 15,000 hours, where the decay structure becomes more obvious.
- Some models that may exhibit a good fit at 6,000 or 15,000 hours may not be realistic.
- Measurement uncertainty in the data could result in poor model fit and extravagant life estimates. To achieve an appropriate estimate, the standard errors on the parameters of the model fits are used to construct a confidence band around the mean prediction. This provides for the representation of a prediction based on a confidence that a specific sample will fall within a certain range around the mean value.
- Measurement repetition, data length, and measurement frequency can support better model fit, though data length is easily the best way to improve model selection (by showing more of the degradation curve).
- In using models fit to only 6,000 hours of a dataset, the model with the best RMSE fit does not consistently estimate the closest actual τ_{70} at the end of the dataset (up to 15,000 hours); this is because at 6,000 hours the product often hasn’t exposed its decay structure (the decay structure tends to change around 5,000 to 6,000 hours, so predictions will not match actual decay rates without sufficient hours of observation).
- It may be reasonable to use RMSE to choose a group of possible models, which may be useful in eliminating poor-fit models.
- Other models may be appropriate to explore, but all (especially those more complicated ones) will be subject to the same restrictions and limitations as those that have already been explored.
- For the most accurate and statistically sound predictions, all instances of a given product type (manufacturer product number) should be representable by one model. If a single model cannot represent all of the individual LEDs of this type, then it should be excluded.
- RMSE may be reasonable for choosing a single best model with data sets much longer than 6,000 hours and most likely well above 10,000 hours. However, for the fast emerging LED industry, longer testing is not believed to be practical, though it is clearly important.

* PRESS statistic: “In statistics, the predicted residual error sum of squares (PRESS) statistic is a form of cross validation used in regression analysis to provide a summary measure of the fit of a model to a sample of observations that were not themselves used to estimate the model. It is calculated as the sums of squares of the prediction residuals for those observations.” (Source: Wikipedia – PRESS statistic. Accessed 2018 Jul 31.)

G.5 References for Annex G

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Annex H – Rationale for Method of Drive Current Interpolation

One of the primary objectives of the TM-21 Working Group (WG) was development of a method that allowed interpolation between different drive currents at a given case temperature of a device under test, as well as simultaneous interpolation between both drive currents and DUT case temperatures.

Three different mathematical models for interpolating the decay rate constant with respect to drive current

were considered: a power law dependence, an exponential dependence, and a linear dependence (see **Table H-1**).

Table H-1. Mathematical Models for Interpolation.

Model	General Equation
Power Law	$\alpha = CI_f^n$
Exponential	$\alpha = A \exp(cI_f)$
Linear	$\alpha = mI_f + b$

The models were evaluated against the data set used by the WG. Parameters for each model were calculated using data sets that included at least 3 different drive current values at the same DUT case temperature. No data sets included more than 4 different drive current values at the same DUT case temperature.

Since all models had the same number of parameters, the sum of squared errors of prediction (SSE) was used as the selection criteria. Based on this criterion, the linear model yielded the best overall performance, followed by the power law model and then the exponential model. The average values of the residuals for the power law and exponential models were, respectively, about 25% and 80% higher than for the linear model.

For simultaneous (two-dimensional) interpolation, the WG evaluated different methods and settled on a sequence of drive current interpolation followed by temperature interpolation. This reduces the problem to sequence of one-dimensional interpolations and avoids the algebraic complexities of simultaneously finding the values of four different parameters.

Changing the order of interpolation results in slight differences of about 1% to 3% in the calculated values of flux maintenance lifetime. In order to avoid slight discrepancies, the WG fixed the order of interpolation as current interpolation followed by temperature interpolation.

In some data sets there were observable changes in the behavior over large changes in drive current that could

result in significant over-prediction of flux maintenance lifetimes. In order to minimize these errors, the WG instituted limits on the maximum change in drive current between interpolation points.

As discussed in **Section 8**, this requires that any interpolation point be bounded by four data points, forming a quadrilateral. When plotted using temperature as the abscissa and current as the ordinate, the left and right sides of the quadrilateral need to be vertical. Examples of allowed regions for simultaneous current and temperature interpolation are shown in **Figures H-1 and H-2**.

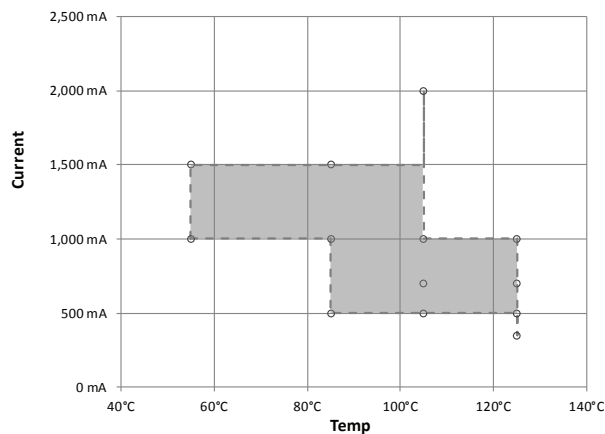


Figure H-1. Example of an allowed two-dimensional interpolation current and temperature interpolation region. Open circles indicate temperature and current conditions for ANSI/IES LM-80-15 test data. Two-dimensional current and temperature interpolation is allowed in the gray region.

The situation can be more easily thought of as performing current interpolation at different two different temperatures to create new data points at the same current but different temperatures, followed by temperature interpolation. Because of this situation, if a data point exists with a drive current that matches the interpolation current, it is possible to interpolate along the line between that data point and a line perpendicular to a pair of data points at a different temperature, with drive currents above and below the interpolation current.

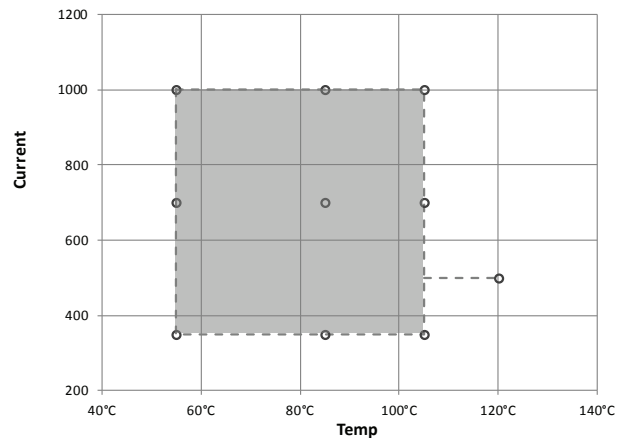


Figure H-2. Example of an allowed two-dimensional interpolation current and temperature interpolation region. Open circles indicate temperature and current conditions for ANSI/IES LM-80-15 test data. Two-dimensional current and temperature interpolation is allowed in the gray region bounded by vertical lines at 55 °C and 105 °C and horizontal lines at 350 mA and 1,000 mA. Interpolation is also allowed along the line extending from 105 °C and 500 mA to 120 °C and 500 mA.

Annex I – Analysis of Uncertainty of Calculated Values of Alpha

It has been noted that in some cases a negative value of α is calculated when using the curve-fit procedures outlined in **Section 5.2.4**. This may be due to a short-term initial increase in flux parameters seen in many LED packages, or uncertainty (noise) in the measurements at one or more time intervals. Regardless of the reasons, a negative value of α implies that the flux parameter will increase monotonically with time. This is unrealistic. In ANSI/IES TM-21-11, this situation was addressed by assuming a constant value for flux maintenance out to the extrapolation limits for the data set.

Since the only realistic situation is described by $\alpha > 0$, the WG conceived the concept of a minimum value of α . This minimum value would be used in place of a constant level of flux maintenance. Uncertainty exists for each measurement and calibration made to a measurement system during ANSI/IES LM-80 testing. These uncertainties cannot be eliminated, only reduced. The uncertainties correspond to a minimum value of α

below which it cannot be meaningfully differentiated from zero.

Annex C describes the calculations for performing a least-squares fit of data. This allows calculations of the parameters for the linear fit, namely the slope and the intercept. These calculations can be extended to calculate the standard error (SE) of the fit according to:

$$SE = \sqrt{\frac{\sum (y_i - y_{i,est})^2}{n - 2}}$$

This in turn allows calculation of the standard error of the slope (SE_{slope}) according to:

$$SE_{slope} = SE \sqrt{\frac{1}{\sum (x_i - x_{ave})^2}}$$

and the standard error of the intercept ($SE_{intercept}$) according to:

$$SE_{intercept} = SE \sqrt{\frac{\sum x_i^2}{n \sum x_i^2 - (\sum x_i)^2}}$$

and then multiplying these standard errors by the Student's t -distribution and the associated confidence interval, in this case 95%.

New ANSI/IES LM-80-15 data sets were available for the WG to analyze, which included a total of 28 different LED products and 211 different combinations of drive current and case temperature. Over half of these datasets extended to at least 9,000 hours. These datasets were then used to evaluate the confidence limits for α .

A first evaluation was conducted by treating each LED product, drive current and case temperature as a 6,000-hour dataset. This analysis gave a median confidence interval for α of $\pm 2.36 \times 10^{-6}$. The analysis was repeated at longer time intervals extending out to 9,000 hours, which provided a total of 618 data points. The median confidence interval of α for this extended dataset was 2.04×10^{-6} (see **Figure I-1**).

An alternative analysis was conducted that focused on datasets and time intervals that resulted in $\alpha < 0$. A total of 70 examples of $\alpha < 0$ were found in the extended dataset by analyzing the time and drive current conditions as 6,000-, 7,000-, 8,000- and 9,000-hour datasets.

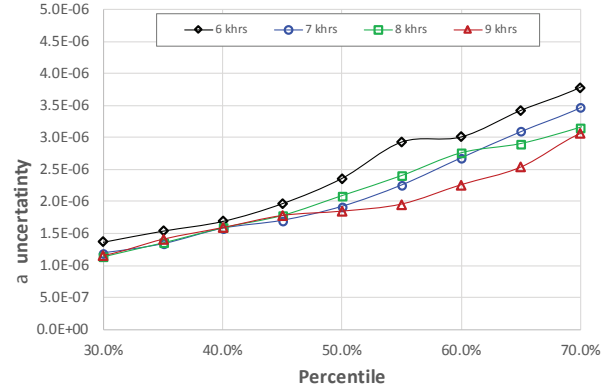


Figure I-1. Uncertainty of α as a function of ANSI/IES LM-80-15 test duration.

In each instance, the magnitude of the confidence limit of α exceeded the absolute value of α (see **Figure I-2**). In other words, the confidence limit included the region of $\alpha > 0$. This supported the hypothesis of a minimum (positive) value of α . The median value of the upper end of the confidence limit was 2.44×10^{-6} .

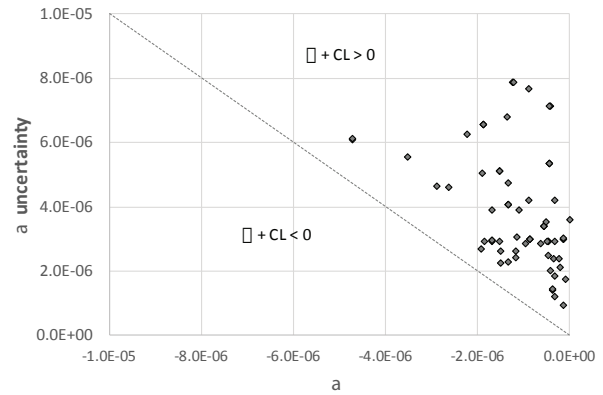


Figure I-2. Uncertainty in calculated value of α as a function of α for $\alpha < 0$.

Both approaches gave similar estimates for the minimum value of α . The WG had some concerns that using two significant digits might convey a false level of precision in the analysis and determined that a minimum value of $\alpha = 2.0 \times 10^{-6}$ was appropriate. Depending on the value of B for the data set, this would correspond to theoretical flux maintenance lifetimes (τ_{70}) between about 152,000 hours and about 203,000 hours. It is noted that this could potentially result in a conflict

with extrapolation limits, but only for ANSI/IES LM-80 datasets that extend past 25,000 hours, which was deemed an unrealistic situation.

Modifications to the value of α without also modifying B will result in a step change in flux maintenance at the transition between the experimentally measured data and the projection. In order to eliminate this, whenever the value of α is modified the value of B should be recalculated using the last experimental data point.

Annex J – Interpolation of Value For B

Initially it was assumed that the calculated value of B would be approximately 1.0 and that there would be little difference between values of B at different test conditions. This led to the adoption of using the geometric mean for the value of B when interpolating flux maintenance life between different test conditions. For LEDs that exhibit little change in flux maintenance over long periods, this is a reasonable approach since the data points used for the curve-fit procedure of **Section 5.2.4** approximate a horizontal line. This naturally leads to B approximately equal to 1. In addition, small changes in the slope of the data points at different case temperature and drive current conditions results in a very small change in B .

When the WG evaluated the new ANSI/IES LM-80-15 data sets, it became evident that use of a geometric mean for B could lead to artifacts in flux maintenance predictions when interpolating between conditions. This occurs because the value of B changes in a stepwise manner between the interpolation conditions. It is important to note that these artifacts were observed for LEDs that would reach τ_{70} before the 6x extrapolation limit.

Examples in this annex are shown as calculations for 70% flux maintenance (τ_{70}). The same effects will be present to at other flux maintenance levels.

An example from the ANSI/IES LM-80-15 datasets evaluated by the WG is shown in **Figure J-1** (the data

are provided in **Tables J-1** and **J-2**). Since the datasets extended to 9,000 hours, the time interval from 4,000 to 9,000 hours, inclusive, was used to calculate α and B . The calculated parameters were then used to extrapolate theoretical behavior back to time = 0.

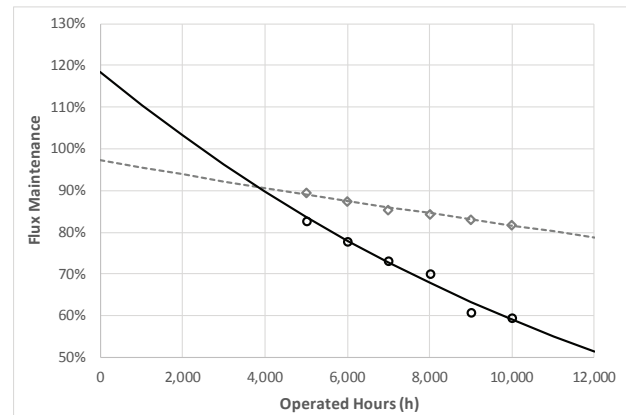


Figure J-1. Flux maintenance data for an LED at the same drive current for case temperatures of 105°C (◊) and 120°C (○). Dashed and solid lines are modeled behaviors based on calculated values of α and B for each data set.

Table J-1. ANSI/IES LM-80-15 Data for B Interpolation Example

Time (h)	Flux, 320 mA		ln(Flux), 320 mA	
	105 °C	120 °C	105 °C	120 °C
0	1.00000	1.00000	0.00000	0.00000
1,000	0.96993	0.94828	-0.03053	-0.05310
2,000	0.95029	0.91885	-0.05098	-0.08463
3,000	0.92698	0.88681	-0.07582	-0.12013
4,000	0.90884	0.85443	-0.09558	-0.15732
5,000	0.89454	0.82954	-0.11145	-0.18688
6,000	0.87553	0.77993	-0.13293	-0.24854
7,000	0.85485	0.73314	-0.15683	-0.31042
8,000	0.84269	0.70109	-0.17115	-0.35512
9,000	0.83236	0.60876	-0.18349	-0.49634

Table J-2. IES TM-21 Parameters for ANSI/IES LM-80-15 Data in Table J-1

Metric	105 °C	120 °C
α	1.745E-05	6.965E-05
B	0.9719	1.1854
τ_{70} (h)	18,805	7,562

Table note: τ_{70} for the 120 °C condition is reached experimentally in the data set and would have a reported value of 8,012 hours.

For this dataset, the values of B at 105 °C and 120 °C are 0.9719 and 1.1854, respectively. In this instance, using the geometric mean for interpolating the value of B between temperatures results in a large step change of about 10% at temperatures immediately adjacent (e.g., ± 0.1 °C) to the ANSI/IES LM-80-15 test temperatures of 105 °C and 120 °C. By calculating τ_{70} as a function of temperature the concern about using the geometric mean for interpolation becomes immediately obvious.

As seen in **Figure J-2**, a slight increase in temperature from 105 °C results in an immediate 30% increase in the calculated value of τ_{70} . In contrast, a slight decrease in temperature from 120 °C results in an immediate almost 20% decrease in the calculated value of τ_{70} .

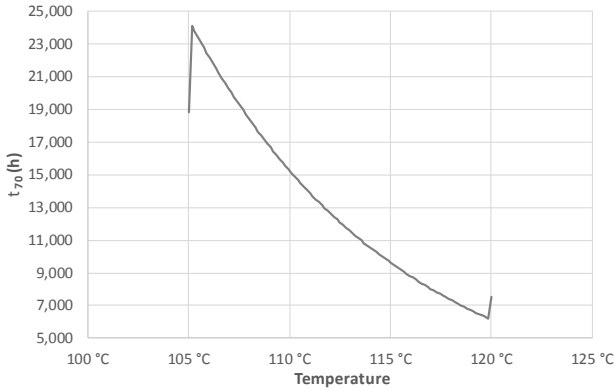


Figure J-2. τ_{70} as a function of temperature for example LED data set.

This behavior is contradictory to expectations and nonsensical. Logically, the calculated value of L_{70} should vary continuously between data points. This can be accomplished by using a linear interpolation scheme for B . This is done using the following equations:

$$B_i = m_B T_{A,i} + b_B,$$

where m_B is the slope:

$$m_B = \frac{B_1 - B_2}{T_{A,1} - T_{A,2}}$$

and b_B is the intercept:

$$b_B = B_1 - m_B T_{A,1}$$

The results of using this linear interpolation scheme for the value of B are compared to the geometric mean, as shown in **Figure J-3**. It can be seen that linear interpolation results in a calculated value of Φ_{70} that changes smoothly between temperatures.

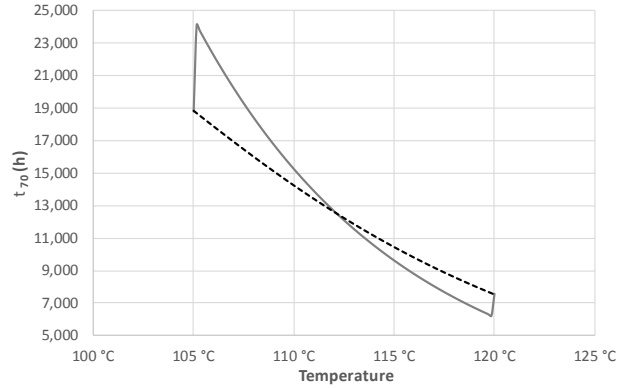


Figure J-3. τ_{70} as a function of temperature for the example LED dataset. The solid line represents interpolation using the geometric mean for B ; the dashed line represents interpolation using a linear function of B .

The WG considered using $1/T_{abs}$ as the interpolation variable but rejected this approach due to concerns that it could associate the interpolation of B with the Arrhenius interpolation used for α . In addition, the calculated values of flux maintenance life (i.e., τ_{70}) using a linear function of temperature and a linear function of $1/T_{abs}$ give results that differed by less than 0.5%.

A similar linear interpolation of B between drive currents was also adopted for consistency and to avoid other potential sources of step changes in calculated values for flux maintenance life (i.e. τ_{70}). This is done using the following equations:

$$B_i = m_B I_i + b_B$$

where m_B is the slope:

$$m_B = \frac{B_1 - B_2}{I_1 - I_2}$$

and b_B is the intercept:

$$b_B = B_1 - m_B I_1$$

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

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2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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